YAKEEN

NEET 2.0 2027

Physics | Revision-01

Basic Maths and Calculus

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Todays Goal

Basic math = 1

* Basic calculation algebra

* log. ✓

* Trigonometry ✓

* A.P | G.P

* Graph. ✓

$|A| = |B| = A$

$R = 2R\cos(\theta/2)$

Basic calculation

= Raftaar नहीं !

$2^0 = 1$

$2^1 = 2$

$2^2 = 2 \times 2 = 4$

$2^3 = 2 \times 2 \times 2 = 8$

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

$2^5 = 16 \times 2 = 32$

$2^6 = 2^5 \times 2^1 = 32 \times 2 = 64$

$2^7 = 2^5 \times 2^2 = 32 \times 4 = 128$

$2^8 = 2^7 \times 2 = 125 \times 2 = 256$

$x^2 \times x^3 = x^5$

$2^9 = 2^8 \times 2 = 512$

$2^{10} = 1024$

$3^0 = 1$

$3^1 = 3$

$3^2 = 3 \times 3 = 9$

$3^3 = 3 \times 3 \times 3 = 27$

$3^4 = 3^3 \times 3 = 81$

$3^5 = 81 \times 3 = 243$

$3^6 = 243 \times 3 = 750$ (729, -21)

$3^7 = 729 \times 3 = 2187$

$3^8 = 2187 \times 3 =$ ——

$4^0 = 1$

$10^0 = 1$

$(x)^0 = 1$

$x^2 = x \times x$

$x^3 = x \times x \times x$

$x^5 = x \times x \times x \times x \times x$

$x^{-2} = \frac{1}{x^2} = \frac{1}{x \times x}$

$\frac{1}{x^{-4}} = x^{+4} =$

$4^1 = 4$

$4^2 = 16$

$4^3 = 4 \times 4 \times 4 = 64$

$4^4 = 64 \times 4 = 256$

$4^4 = 4^2 \times 4^2 = 256$

$4^5 = 256 \times 2 \times 2$

$(2^2)^5 = 2^{10} = 1024$ $= 1024$

$4^6 = 4^3 \times 4^3$

$= 2^6 \times 2^6$

$= 64 \times 64$

$5^0 = 0$

$5^1 = 5$

$5^2 = 25$

$5^3 = 125$

$5^4 = 125 \times 5$

$= 625$

$6^1 = 6$

$6^2 = 36$

$6^3 = 36 \times 6$

$= 216$

$1^2 = 1$
$2^2 = 4$
$3^2 = 9$
$4^2 = 16$
$5^2 = 25$
$6^2 = 36$
$7^2 = 49$
$8^2 = 64$
$9^2 = 81$
$10^2 = 100$

$11^2 = 121$
$12^2 = 144$
$13^2 = 169$
$14^2 = 196$
$15^2 = 225$
$16^2 = 256$
$17^2 = 289$
$18^2 = 324$
$19^2 = 361$
$20^2 = 400$

(13) 13 × 13 = 39 + 130 = 169

(14) 14 × 14 = 56 + 140 = 196

18 × 18 = 126 + 180 = 306 + 18 = 324

17 × 17 = 119 + 170 = 289

16 × 16 = 96 + 160 = 256

(18) 18 × 18 = 144 + 180 = 324

19 × 19 = 171 + 190 = 361

$1^3 = 1$

$2^3 = 2 \times 2 \times 2 = 8$

$3^3 = 27$

$4^3 = 64$

$5^3 = 125$

$6^3 = 216$

$7^3 = 343 \quad (49 \times 7)$

$8^3 = 64 \times 8 = 512$ (350)

$9^3 = 81 \times 8 = 648$ → $\frac{81}{8}$ / $\frac{648}{}$

$10^3 = 1000.$

Maha-Basic-1

$0.1 = \frac{1}{10}$ $\quad 0.01 = \frac{1}{100} = \quad 0.002 = \frac{2}{1000}$

$\frac{1}{10} = 0.1$ ✓

$\Rightarrow \frac{1}{100} = 0.01 = 10^{-2}$ ✓

$\frac{1}{1000} = \frac{1}{10^3} = 10^{-3} = 0.001$

$10^{-4} = \frac{1}{10^4} = 0.0001$

QUESTION

$0.4 = 0.40 = 0.400$



Maha-Basic-1

→ $0.2 = \frac{2}{10} = \frac{1}{5} = 0.2$

$0.4 = \frac{4}{10} = \frac{2}{5} = 0.4$ ✗ No Rath

\# $0.50 = \frac{50}{100} = \frac{1}{2} = 0.5$

$0.60 = \frac{60}{100} = \frac{3}{5} = 0.6$

\# $0.75 = \frac{75}{100} = \frac{3}{4}$

$0.33 = \frac{33}{100}$ ≠ $\frac{1}{3}$

MULTIPLICATION AND DIVISION

$0.5 = \frac{1}{2}$

$0.6 = \frac{3}{5}$

$0.4 = \frac{2}{5}$

\# $0.6666 = \frac{66}{100}\,{}^{2}_{3} \approx \frac{2}{3}$

$0.71 = \frac{71}{100}$

\# $1.33 = \frac{133}{100}\,\frac{4}{3} \approx \frac{4}{3}$

$0.75 = \frac{3}{4}$

$0.33 = \frac{1}{3}$

0.4×25 $= \frac{2}{\cancel{10}_{5}} \times \cancel{25}^{5} = 10$

0.75×25 $= \frac{\cancel{75}^{3}}{\cancel{100}_{4}} \times 25 = \frac{75}{4}$

0.67×12 $= \frac{2}{\cancel{3}} \times \cancel{12}^{4} = 8$

0.33×15 $= \frac{1}{\cancel{3}} \times \cancel{15} = 5$

$\frac{1.33}{0.25}$ $= \frac{4}{3 \times \frac{1}{4}} = \frac{16}{3}$

$0.66 \simeq 0.67 = \frac{2}{3}$

Maha-Basic-1

$0.66 = \frac{66}{100}$

$0.25 = \frac{25}{100}$

$1.33 = \frac{133}{100}$

$1.5 = \frac{15}{10} = \frac{3}{2}$

$4.33 = \left(\frac{433}{100}\right) = (3 + 1.33) = \left(3 + \frac{4}{3}\right)$

$2.5 = \frac{5}{2}$

- $7.4 \times 25 =$ $(7+0.4)\times 25 = 175 + \frac{2}{5}\times 25 = 185$
- $2.66 \times 15 =$ $(2+0.66)\times 15 = 15\times 2 + \frac{2}{3}\times 15^{5} = 40$
- $8.75 \times 8 =$ $(8+0.75)\times 8 = 64 + \frac{3}{4}\times 8^{2} = 64+6 = 70$
- $6 \times 4.33 =$ $6\left(4+\frac{1}{3}\right) = 24+2 = 26$
- $25 \times 0.6 =$ $2\frac{5}{5}\times \frac{6^{3}}{10_{5}} = 15$

$$\begin{array}{r} 2.66 \\ 15 \\ \hline \end{array}$$

$$\begin{array}{r} 25 \\ \times\ 7.4 \\ \hline 185 \end{array}$$

Maha-Basic-1

$12 \times 0.67 =$

$15 \times 0.67 =$

$0.125 \times 24 =$

[redacted]

Also $0.66 = 0.67 \approx \frac{2}{3}$

$$\sqrt{0.04} = \sqrt{\frac{4}{100}} = \frac{2}{10} = 0.2$$

$$\sqrt{0.64} = \sqrt{\frac{64}{100}} = \frac{8}{10} = 0.8$$

$$\sqrt{0.49} = \sqrt{\frac{49}{100}} = 0.7$$

$$\sqrt{0.016} = \sqrt{\frac{16}{1000}} = \frac{4}{10\sqrt{10}}$$

✓ $(0.4)^2 = \left(\frac{4}{10}\right)^2 = \frac{16}{100} = 0.16$

✓ $(0.5)^2 = \frac{25}{100} = 0.25$

$(1.33)^2 = \left(\frac{133}{100}\right)^2 = \frac{16}{9}$

$\# (1.33)^2 \times \frac{18}{4} = \frac{16}{9} \times \frac{18}{4} = 8$

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x & x^2

- $x^2 > x$ (if $x > 1$)
- $x^2 = x$ if $x = 1$
- $x^2 < x$ if $x < 1$

$1\,fm = 10^{-15}\,m$

$1\,PC = 10^{-12}\,C$ (Pico)

$1\,\eta c = 10^{-9}\,C$ (nano)

$1\,\mu F = 10^{-6}\,F$ (micro)

$1\,mm = 10^{-3}\,m$ (milli)

$1\,KJ = 10^{3}\,J$ (Kilo)

$1\,MW = 10^{6}\,W$ (Mega)

$1\,GN = 10^{9}\,N$ (Giga)

What is infinity ?

Which is non-zero

- If Power of any number is zero then result will be one.

$x^\circ = 1$

$8^\circ = 1$

$(MR)^\circ = 1$

$(Ramlal)^\circ = 1$

$0^\circ = X$

What is infinity ?

- Is one divided by zero is Infinite??

$\frac{1}{0} = \infty$

$\frac{1}{1} = 1$

$\frac{1}{10} = 0.1$

$\frac{1}{100} = 0.01$

$\frac{1}{1000} = 0.001$

$\frac{1}{10000} = 0.0001$

$\frac{1}{1000000} = 0.000001$

$\frac{1}{\infty} = 0$

$\frac{1}{0} = \infty$

1m is infinitly large in compaire to Radius of atom.

1m is infinitly small in Compair to Radius of earth

RULE OF POWER (EXPONENT)

1. If power of any non-zero number is zero then result will be 1.

$8^0 = 1$ $10^0 = 1$

2. Negative Property of exponent (x is non-zero number)

$x^n = \frac{1}{x^{-n}}$ $\frac{1}{x^n} = x^{-n}$

$2^6 = \frac{1}{2^{-6}}$ $10^3 = \frac{1}{10^{-3}}$ $\frac{1}{10^4} = 10^{-4}$ $\frac{1}{10^{-6}} = 10^{+4}$ $10^{-2} = \frac{1}{10^2}$

3. Product property of exponent

$$x^n \cdot x^m = x^{n+m}$$

$2^2 \times 2^6 = 2^8$

4. Division property of exponent

$$\frac{x^n}{x^m} = x^{n-m}$$

$$\frac{10^6}{10^3} = 10^{6-3} = 10^3$$

5. Power of a power :

$$(x^n)^m = x^{n \times m}$$

$(2^2)^3 = 2^5$ ✗

$(2^{2+3}) = 2^6$

$(2^2)^3 = 4^3 = 64$

$2^6 = 64$

$(3^2)^2 = 3^4$

6. Fractional exponent :

$$(x)^{3/2} = (x^3)^{1/2} = \sqrt[2]{x^3}$$

$(2^4)^3 = 2^{12}$

$x^{1/2} = \sqrt{x}$

$x^{1/3} = \sqrt[3]{x}$

$x^{3/2} = (x)^{\frac{1}{2} \times 3} = (x^3)^{\times \frac{1}{2}}$

or

QUESTION

Find value of $10^2 + 10^3$??

A 10^5 ✗

B 10^6

C 1.1×10^3

D 101

$10^2 + 10^3 = 100 + 1000$

$= \frac{1100}{1000} \times 1000$

$= 1.1 \times 10^3$

$10^2 \times 10^3 = 10^5$

$4.37 = 437 \times 10^2$ ✗

$4.37 = \frac{437}{100} = 437 \times 10^{-2}$

$0.378 = 378 \times 10^{-3}$

$3.789 \times 10^{-2} = 3789 \times 10^{-5}$

$2.78 \times 10^4 = \frac{278}{100} \times 10^{4 \times 10^2}$

$= 278 \times 10^2$

$(8)^{2/3} = (8)^{\frac{2}{3}} = (2^3)^{\frac{2}{3}} = 2^{3\times\frac{2}{3}} = 4$

$(4)^{5/2} = 2^{2\times\frac{5}{2}} = 2^5 = 32$ ✓

$(16)^{3/2} = (4^2)^{\times\frac{3}{2}} = 4^3 = 64$

$\sqrt{8} = \sqrt{4\times2} = 2\sqrt{2}$

$\sqrt[3]{8} = (8)^{1/3} = (2^3)^{1/3} = 2$

$\sqrt{2} = 1.41$

$\sqrt{3} = 1.71$ $\sqrt{5} = 2.4$

$2^3 = 8$

$\sqrt{4} = (4)^{1/2}$

$\sqrt[3]{4} = (4)^{1/3}$

$y = x^2$

$y^{\frac{1}{2}} = x$

$\sqrt{y} = x$

$y = x^{1/3}$

$x = y^3$

(*) $y = x^{3/2}$

$x = y^{\frac{2}{3}}$

$x = y^{-\frac{2}{3}}$

$y^{-2} = x^3$

$y = x^{-\frac{3}{2}}$

QUESTION

Find value of x for given expression ??

$$27 = (5 + x^2)^{3/2}$$

$$27 = (5+x^2)^{3/2}$$

$$(27)^{2/3} = 5+x^2$$

$$(3^3)^{\frac{2}{3}} = 5+x^2$$

$$9 = 5+x^2$$

$$9-5 = x^2$$

$$x^2 = 4$$

$$x = \sqrt{4} = 2$$

QUESTION

$\pi = 3.14 = \frac{22}{7}$, $\pi^2 = 10$ Find value of $0.4\ \pi^2$.

$0.4(\pi^2)$

$\frac{4}{10} \times 10 = 4$

$e = 2.71$

QUESTION

Find value of $3^4 + 2^5$?

$\Rightarrow 81 + 32$

$\Rightarrow 113$

$3^2 = 9$

$3^3 = 27$

$3^4 = 81$

QUESTION

Find value of $4^5 + 4^5$?

$2(4^5)$

$4^5 + 4^5 = 4^5(2)$

$= (2^2)^{x5} \times 2$

$= 2^{10} \times 2$

$= 1024 \times 2$

$= 2048$ Ans.

QUESTION

Find value of $16^{-1/4} + 4^{-2}$?

$$16^{-\frac{1}{4}} + 4^{-2} = \frac{1}{16^{+\frac{1}{4}}} + \frac{1}{4^2}$$

$$= \frac{1}{2} + \frac{1}{16}$$

$$= \frac{8+1}{16} = \frac{9}{16}$$

$$\# (16)^{1/4} = (2^4)^{1/4} = 2$$

$2^2 = 2$
$2^3 = 8$
$2^4 = 16$

$$\frac{1}{a} + \frac{1}{b} = \frac{b+a}{ab}$$

$$\frac{1}{2} + \frac{1}{4} = \frac{2+1}{4} = \frac{3}{4}$$

$$\frac{1}{3} + \frac{1}{4} = \frac{4+3}{12} = \frac{7}{12}$$

$$\frac{1}{2} + \frac{1}{8} + \frac{1}{4} = \frac{4+1+2}{8}$$

$$= \frac{7}{8}$$

QUESTION

Find value of n; if $\frac{2}{n} = 4 + \frac{6}{7}$.

$$\frac{2}{n} = \frac{4}{1} + \frac{6}{7}$$

$$\frac{2}{n} = \frac{28 + 6}{7}$$

$$\frac{2}{n} = \frac{34}{7}$$

$$n = \frac{7}{17}$$

QUESTION

If $2^8 = (16)^n$ then find the value of n.

$2^8 = 16^n$

$2^8 = (2^4)^n$

$2^8 = 2^{4n}$

$4n = 8$

$n = 2$

- $\frac{2}{\sqrt{2}} = \frac{\sqrt{2} \times \sqrt{2}}{\sqrt{2}} = \sqrt{2}$

$\sqrt{3} \times \sqrt{3} = 3$

$\sqrt{4} \times \sqrt{4} = 4$

$\sqrt{2} \times \sqrt{2} = 2$

- $\frac{10^{-4}}{2} = \frac{10^{-4}}{2} \times \frac{10}{10}^{5}$

$= 5 \times 10^{-5}$

- $\frac{10^3}{5} = \frac{\overset{2}{10} \times 10^2}{5} = 2 \times 10^2$

Maha-Basic-2

Find x in given expansion:

$(x)^{\frac{3}{3}} = 4$

$(x)^{1/3} = 4$

$\Rightarrow x = 4^3 = 64.$

$(x)^{\frac{2}{3}} = 9$

$\Rightarrow (x)^{2/3} = 3^2$

$x = 3^{\frac{2 \times 3}{2}} = 27$

Maha-Basic-2

Find x in given expansion:

$(x)^{\frac{3}{4}} = 27$

H/W

$(x)^{\frac{3}{2}} = 8$

H/W

Maha-Basic-3

$$\frac{4}{(1/x)} = 3 \quad \Rightarrow \quad \frac{4}{1/x} = 3$$

$$4x = 3$$

$$x = \frac{3}{4}$$

$$\frac{y}{(1/3)} = 4$$

$$y = 4 \times \frac{1}{3} = \frac{4}{3}$$

$$\frac{x}{(1/y)} = 8$$

$$yx = 8$$

$$\frac{1/x}{y} = \frac{1}{xy}$$

$$\frac{1/y}{y/x} = \frac{1}{y} \times \frac{x}{1} = \frac{x}{y}$$

- $4^{x/2} = 8$, find x

$(4)^{\frac{x}{2}} = 8$

$2^{2 \times \frac{x}{2}} = 2^3$

$x = 3$ Ans.

- $10^{x/3} = 10^5$, find x

$\frac{x}{3} = 5 \quad x = 15$

- $\frac{10^7}{10^{x/2}} = 10^6$, find x

$\frac{10^7}{10^6} = 10^{x/2}$

$10 = 10^{x/2} \quad \frac{x}{2} = 1 \quad x = 2$

- $x^{-\frac{2}{5}} = \frac{1}{9}$

$x^{-\frac{2}{5}} = \frac{1}{9}$

- $(125)^{2/3} =$ $(5^3)^{\frac{2}{3}} = 25$
- $(216)^{2/3} =$
- $(64)^{2/3} =$

QUESTION (Level up)

1. If $3^{6-x} = 27$, find value of x.

Soln

$3^{6-x} = 27$

$3^{6-x} = 3^3$

$6 - x = 3$

$x = 6 - 3 = 3$

2. If $2^{y-4} = 64$, find value of y.

$y = 10$ Ans

QUESTION (Level up)

3. If $x = \frac{d}{\sqrt{3}-1}$. Simplify this.

$(a+b)(a-b) = a^2 - b^2$

$$x = \frac{d}{(\sqrt{3}-1)} \times \left(\frac{\sqrt{3}+1}{\sqrt{3}+1}\right)$$

$$x = \frac{d\,(\sqrt{3}+1)}{(3-1)}$$

$$\boxed{x = \frac{d(\sqrt{3}+1)}{2}}$$ R

QUESTION (Level up)

4. If $(x-4)^{2/3} = 4$. Find x.

Soln $(x-4)^{\frac{2}{3}} = 4$

$x-4 = (2)^{2 \times \frac{3}{2}}$

$x = 4+8 = 12$

QUESTION (Level up)

5. If $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$. Find x.

$$\frac{a}{b} = \left(\frac{b}{a}\right)^{-1}$$

$$\left(\frac{a}{b}\right)^{x-1} = \left(\frac{a}{b}\right)^{3-x}$$

$$x-1 = 3-x$$

$$2x = 4$$

$$x = 2$$

$$\frac{x^2}{y^2} = \left(\frac{x}{y}\right)^2$$

$$\frac{1y^{-2}}{x^{-2}} = \left(\frac{y}{x}\right)^{-2}$$

QUESTION

$(2^2)^3 = ?$

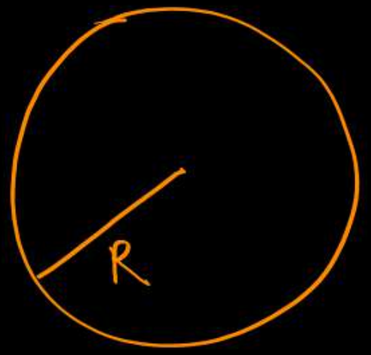
$\Rightarrow 2^6$

A 10

B 12

C 64

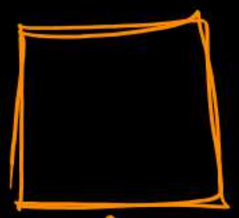
D 32



$\Rightarrow \pi R^2$

$D = 2R$

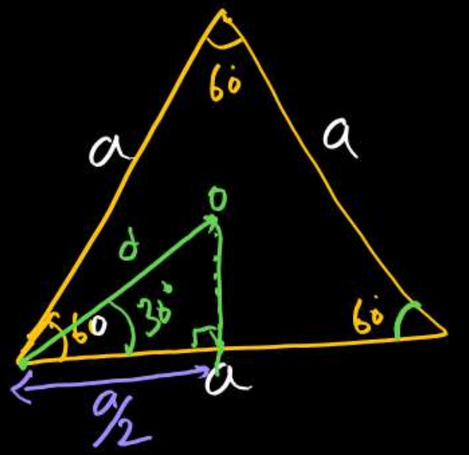
$\text{circumf} = 2\pi R$



$\text{Perim} = 4l \quad A = l^2$



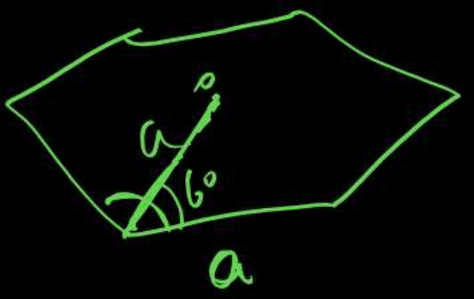
$\text{Circ} = 2b + 2a \quad (a \times b) = \text{Area}$



$\cos 30^\circ = \frac{a}{2 \times d}$

$\frac{\sqrt{3}}{2} = \frac{a}{2d}$

$d = \frac{a}{\sqrt{3}}$



$2^1 = 2$

$2^2 = 4$

2 ka power 3 = 8 → $2^3 = 8$

$2^4 = 16$ = 2 ka power 4 ⇒ 16

$2^5 = 32$

$2^6 = 64$

$3^2 = 9$

$3^3 = 27$

$3^4 = 81$

$4^2 = 16$ $\quad 5^2 = 25$

$4^3 = 64$ $\quad 5^3 = 125$

घुमा के लिखो →

$2^3 = 8$ → Power, Base = 2, Result = 8

Base को as it is रहने दो, Power ⇄ Result को interchange कर दो

log is a concept of ⇒

$\log_2 8 = 3$

$\log_4 16 = 2$ (Base)

$\log_3 9 = 2$ $\quad \log_1 1 = 1$

$\log_2 4 = 2$

Concept of log :-

$\log_2^8 = 3$

$\log_2^4 = 2$

$\log_4^{64} = 3$

$4^3 = 4 \times 4 \times 4 = 64$

$\log_2^{64} = 6$

$2^6 = 64$

Base

$\log_2^{32} = 5$

$\log_2^{64} = 6$

$\log_5^{125} = 3$

$\log y^x = z$ = log y ka Power x = z

X

$\log y^x = z$

↳ log x on the Base y = z

Logarithmic

Common log.

$\Rightarrow \log_{10} x = \log x$

$\log_5 x$

Base = 5

$\log_2 x$ ←

Natural log

$\log_e x = \ln(x)$

↑ natural

(e) Base ही silent

$\ln 10 = \log_e 10$

Property of log :- 1

Base other than 0 and 1

$$\log_2 1 = 0$$

$$\log_3 1 = 0$$

$$\log_4 1 = 0$$

$$\log_{10} 1 = 0$$

$$\log_e 1 = 0$$

$$\log_1 1 = 1$$

$3^0 = 1$

$\log_3 1 = 0$

$$\log_{10}^{\sin 90^\circ} = 0$$

$$\log_e^{(\sin^2\theta + \csc^2\theta)} = 0$$

$$\log_e^{\tan\theta \cdot \cot\theta} = 0$$

Property-2

$$\log x^x = 1$$

$$\log 10^{10} = 1$$

$$\log 2^2 = 1$$

$$\log e^e = 1$$

$$\Rightarrow \log 6^6 = 1$$

Property-1

$\log_{10} 1 = 0$

$\log_2 1 = 0$

$\log_e 1 = 0$

$\log_6 1 = 0$

Prop-2

$\log_6 6 = 1$

$\log_{10} 10 = 2$

$\log_4 4 = 1$

Prop-3

$\Rightarrow \log_e(xy) = \log_e x + \log_e y$

Prop-4

$\Rightarrow \log_e\left(\frac{x}{y}\right) = \log_e x - \log_e y$

Property-5

$\log_e (x)^n \Rightarrow$ log x ka Power n on the Base

$\log_e (x^n) = n \log_e x$

$\log_{(e^n)} x = \frac{1}{n} \log_e x$

Prop-6

$\log_y x = \frac{\log_e x}{\log_e y}$

Prop-7

$\log_y x = \frac{1}{\log_x y}$

(#) $\log_{10} 1 = 0$

$\log_{10} 2 = 0.30$

$\log_{10} 3 = 0.48$

$\log_{10} 4 = \log_{10}^{(2\times2)} = \log_{10} 2 + \log_{10} 2 = 0.6$

$\log_{10} 5 = \log_{10}^{\left(\frac{10}{2}\right)} = \log_{10} 10 - \log_{10} 2 = 1 - 0.3 = 0.7$

$\log_{10} 6 = \log_{10}^{(3\times2)} = \log_{10} 3 + \log_{10} 2 = .78$

(*) $\log_{10} 7 = 0.84$

$\log_{10} 8 = \log_{10}^{(2)^3} = 3 \times 0.3 = 0.9$

$\log_{10} 9 = \log_{10}^{(3)^2} = 2\log_{10} 3 = 2 \times 0.48 = 0.96$

$\log_{10} 10 = 1.$

$\log_e 10 = 2.303$ (α)

$\log_{10} 12 = \log_{10}^{4\times3} = \log_{10} 4 + \log_{10} 3$

$= 0.6 + 0.48$

$= 1.08$

$\log_{10} 15 = \log_{10}^{5\times3} = \log_{10} 5 - \log_{10} 3$

$= 0.7 - 0.48$

$\log_{10}^{12} = \checkmark$

feel

$* \quad \log_{10}^{100} = \log_{10}^{10^2} = 2 \times 1$

$\log_{10}^{5} = 0.7$

$\log_{10}^{6} = 0.78$

$\log_{10}^{15}$ ✓

$\log_{10}^{14} = \log_{10}^{7 \times 2} = \log_{10}^{7} + \log_{10}^{2}$

$\log_{10}^{0.4} = \log_{10}^{\left(\frac{4}{10}=\frac{2}{5}\right)} = \log_{10}^{2} - \log_{10}^{5} = 0.30 - 0.7$

$= -0.4$

$\log_{10}^{0.6} = \log_{10}^{\left(\frac{6}{10}\right)} = \log_{10}^{6} - \log_{10}^{10} = 0.78 - 1$

$\log_{10}^{2.5} = \log_{10}^{\left(\frac{25}{10}\right)} = \log_{10}^{25} - \log_{10}^{10}$

$\log_{10}^{0.01} = \log_{10}^{10^{-2}} = -2 \times \log_{10}^{10}$

$= -2$

$\log_{10}^{10^{-2}} = \frac{1}{100} = 0.01$

$\log_{0.1}^{0.01} = \log_{10^{-1}}^{10^{-2}} = \frac{-2}{-1} \log_{10}^{10}$

$= 2$

$0.01 = \frac{1}{100} = 10^{-2}$

$$\log_e e^{25} + \log_e e^{4} - \log_e e^{10} = \log_e e^{\left(\frac{25\times4}{10}\right)} = \log_e e^{10} = \underline{2.303}$$

\# $\log_{10}(4\times10^{-4}) = \log_{10}4 + \log_{10}10^{-4}$

$= 0.6 + (-4\times1)$

$= 0.6 - 4$

$= \underline{\underline{-3.4}}$

$\log_{10}(\sin\theta \cdot \operatorname{cosec}\theta) = 0$

$\# \log_{10} 200 - \log_{10} 2 = \log_{10}\left(\frac{200}{2}\right) = \log_{10} 100 = 2$

$\# \log_{100} 200 = \log_{(10)^2} 100\times 2 = \frac{1}{2}\left(\log_{10} 100\times 2\right)$

$= \frac{1}{2}\left(\log_{10} 100 + \log_{10} 2\right)$

$= \frac{1}{2}(2 + 0.3) = \frac{2.3}{2}$

$\log_{10} 200 = \checkmark$

$$\rightarrow \log_{8}{16} = \log_{(2)^3}{(2)^4} = \frac{4}{3}\left(\log_2 2\right) = \frac{4}{3} \checkmark$$

$$\rightarrow \log_{27}{3} = \log_{(3)^3}{3} = \frac{1}{3}\log_3 3 = \frac{1}{3} \checkmark$$

$$\rightarrow \log_{10}{0.001} = \log_{10}{10^{-3}} = -3$$

$$\rightarrow \log_{0.1}{10} = \log_{10^{-1}}{10} = \frac{1}{-1}\log_{10}{10} = -1$$

$$0.1 = \frac{1}{10} = 10^{-1}$$

Rule $\log_y x = \frac{1}{\log_x y}$

$\log_e 10 = 2.303$

$\log_{10} 2 = 0.3$ ✓

$\log_e 2 = \frac{\log_{10} 2}{\log_{10} e} = \frac{\log_{10} 2}{1} \times \log_e 10 = 2.303 \times \log_{10} 2$

MR Rutu

$\frac{1}{\log_{10} e} = \log_e 10 = 2.303$

$\log_e x = \log_{10} x \,(2.303)$

$$\log e^4 = \log_{10} 4 \,(2.303)$$

(A) $\log e^{10} = \log_{10} 10^{1} \times 2.303$

QUESTION

Loudness of sound at a point is 50 dB then find intensity at that point if $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB where $I_0 = 10^{-12}$ w/m²

A 10^{-7} w/m² Ans

B 10^{-5} w/m²

C 10^{8} w/m²

D 10^{7} w/m²

Loudness = 50 dB

Intensity

$L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB

$50 = 10 \log_{10}\left(\frac{I}{I_0}\right)$

$5 = \log_{10}\left(\frac{I}{I_0}\right)$

taking anti-log

$10^5 = \frac{I}{I_0}$

$I = I_0 \times 10^5$

$= 10^{-12} \times 10^5$

$= 10^{-7}$

$\log_{10}\left(\frac{I}{I_0}\right) = 5$

Anti-log

$\log_y z = x$

taking Anti-log (Power ke form me)

$y^x = z$

$y^x = z$

y ka Power x is equal to z

$\log_y z = x$

$\Rightarrow$ y ka Power x equal to z

$\Rightarrow$ log of z on the base y is x

QUESTION

If current flowing through wire is $I = 20\ e^{-\lambda t}$ find the time when current becomes 10 Amp. Where λ is 2.303.

$I = 20\ e^{-\lambda t}$

$10 = 20\ e^{-\lambda t}$

$1 = \frac{2}{e^{\lambda t}}$

$e^{\lambda t} = 2$

taking $\log_e$ both side

$\log_e (e)^{\lambda t} = \log_e 2$

$\lambda t \log_e e = 2.303 \times \log 10^2$

$1 \times 2.303 t = 2.303 \times 0.3$

$t = 0.3$ sec

$\log e^2 = \log_{10}^2 \times 2.303$

Trigonometry

$P = \sqrt{H^2 - B^2}$

$B = \sqrt{H^2 - P^2}$

$P^2 + B^2 = H^2$

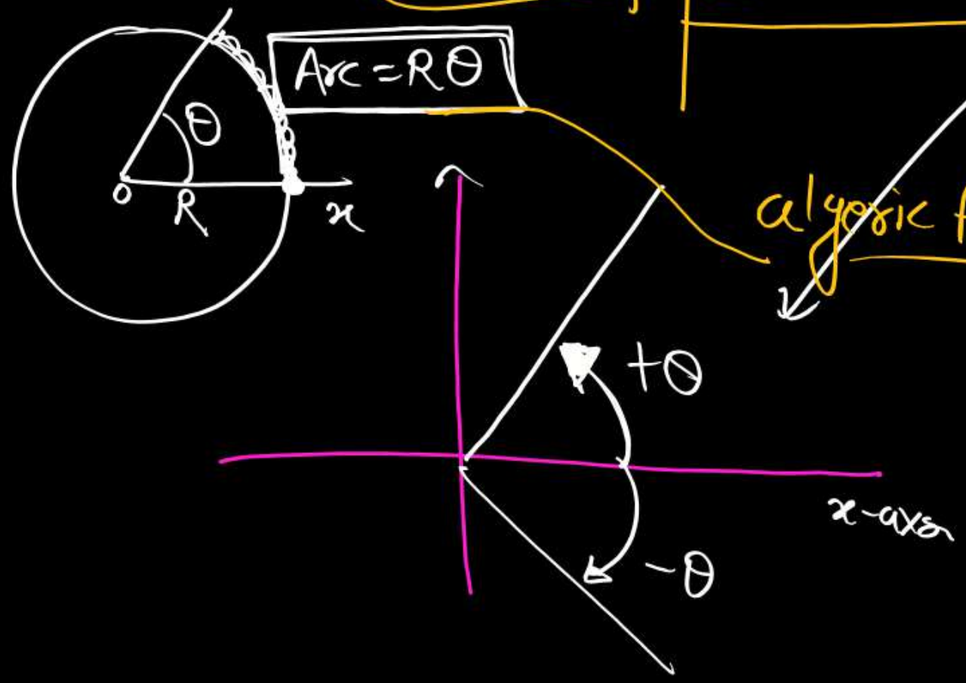
$H = \sqrt{P^2 + B^2}$

On right angle triangle angle always taken B/w Base & hypotenuse

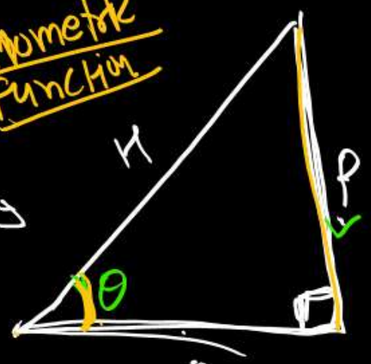
We talk about angle

$\cos\theta \cdot \sec\theta = 1$

$\sin\theta \cdot \operatorname{cosec}\theta = 1$



Trigonometric Function

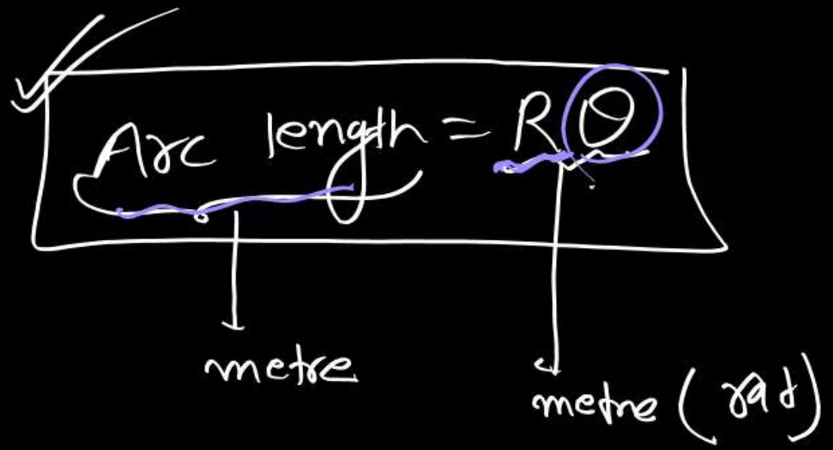


$\sin\theta = \frac{P}{H}$; $\operatorname{cosec} = \frac{H}{P}$

$\cos\theta = \frac{B}{H}$, $\sec = \frac{H}{B}$

$\tan\theta = \frac{P}{B}$; $\cot\theta = \frac{B}{P}$

(4) $\cot\theta \cdot \tan\theta = \frac{B}{P} \times \frac{P}{B} = 1$



Angle, have unit (rad) but does not have dimension.

UNIT OF ANGLE

S.I. units = Radian

Other unit (Practical unit) = degree, minute, second

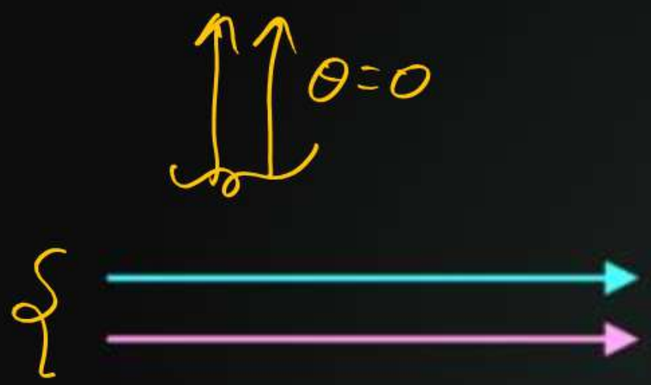
For algebraic function (θ) we always use radian, for trigonometric ratio we can use both degree or radian

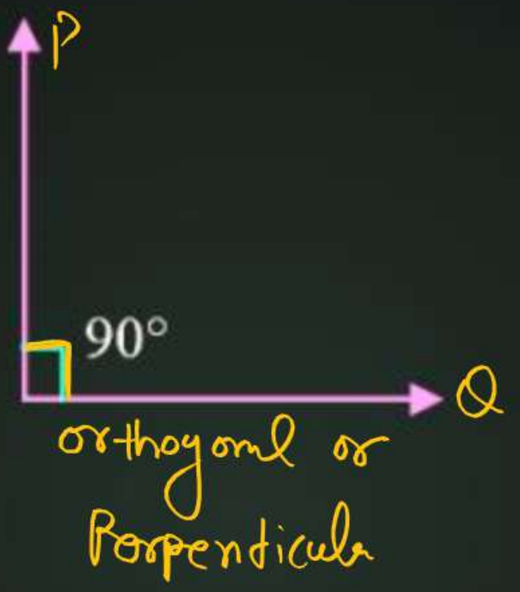
$\sin(60^\circ) =$

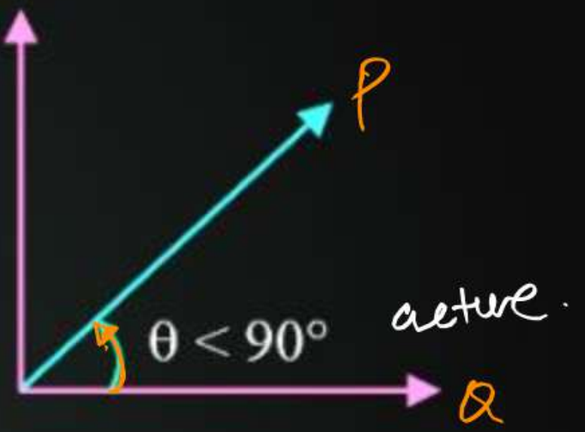
$\sin(60\text{rad}) =$

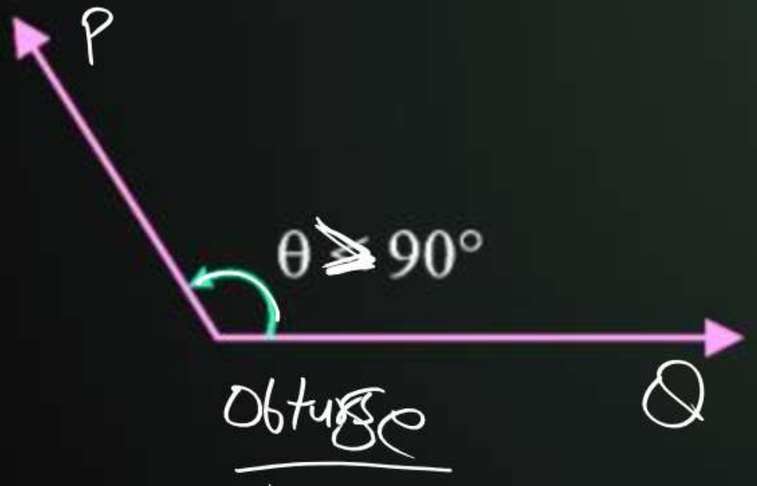
$\text{Arc} = R\,60^\circ$

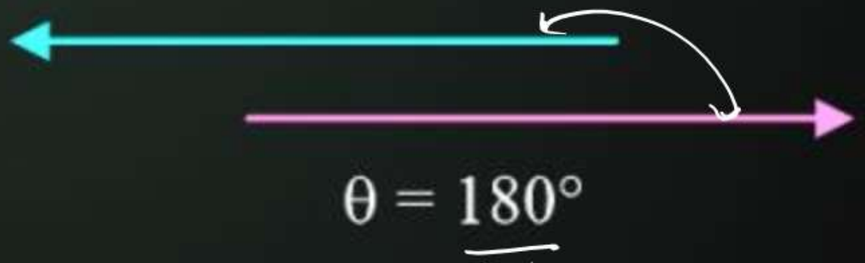
$= R\left(\frac{\pi}{3}\right)$

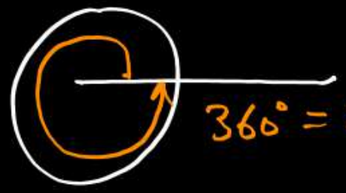

θ=0
Angle θ = 0°


P
90°
Q
orthogonal or
Perpendicular


P
θ < 90°
acute.
Q


P
θ ≥ 90°
Obtuse
Q


θ = 180°



$360^\circ = 2\pi\ rad$

$180^\circ = \pi\ rad$

MR Rafik

$180 \times 1^\circ = \pi\ rad$

$1^\circ = \frac{\pi\ rad}{180}$

$1\ rad = \frac{180^\circ}{\pi}$ ✓✓

180° → out of Phase

QUESTION

$180° = \pi$ rad

Convert following radian into degree

(i) $\frac{\pi}{2}$ rad $= \frac{\pi}{2} \times \frac{180°}{\pi} = 90°$ → MR* $\frac{180°}{2} = 90°$

(ii) $\frac{\pi}{4}$ rad $= \frac{180°}{4} = 45°$

$\frac{\pi}{6}$ rad $= 30°$

(iii) $\frac{\pi}{3}$ rad $= \frac{180°}{3} = 60°$

$\frac{3\pi}{4}$ rad $= 135°$

(iv) $\left[\frac{5\pi}{6} \text{ rad}\right] = \frac{5}{6} \times 180° = 150°$

$\frac{4}{6}\pi$ rad $= \frac{2}{3}\pi$ rad $= 120°$

(v) $\frac{2\pi}{3}$ rad $= \frac{2}{3} \times 180 = 120°$

$\frac{3\pi}{2}$ rad $= 270°$

(vi) $\frac{4\pi}{3}$ rad $= \frac{4}{3} \times 180° = 240°$

QUESTION

$180° = \pi$ rad

Convert following into radian

$180° = \pi$ rad

$\pi = 3.14$

π rad $= 180°$ ✓

(i) $45° = 45 \times \frac{\pi \text{ rad}}{180} = \frac{\pi}{4}$ rad.

(ii) $60° = \frac{3 \times 60°}{3} = \frac{\pi \text{ rad}}{3}$ ✓

(iii) $120° = \frac{2\pi}{3}$ rad ✓

(iv) $30° = \frac{\pi}{6}$ rad

(v) $150° = \frac{5\pi}{6}$ rad

(vi) $90° = \frac{\pi}{2}$ rad

(vii) $\pi^\circ = \pi \times 1° = \pi\left(\frac{\pi \text{ rad}}{180}\right) = \frac{\pi^2 \text{ rad}}{180} = \frac{1}{18}$ rad

(viii) $270° = 3\left(\frac{\pi}{2} \text{ rad}\right)$

(ix) $135° = 3(45°) = \frac{3\pi}{4}$ rad.



QUESTION

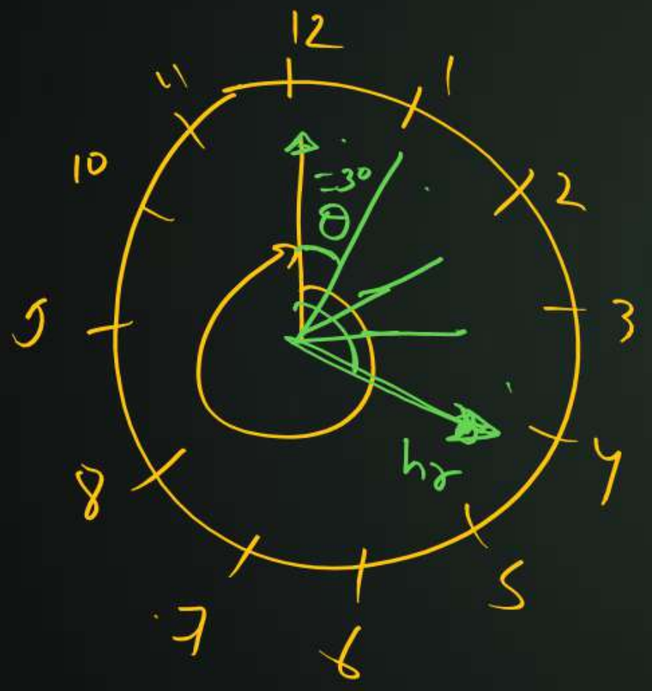
Convert 1 radian into minute ?

$1^\circ = 60 \text{ mint}$

$$1 \text{ rad} = \frac{180^\circ}{\pi} = \frac{180 \times 60 \text{ m}}{\pi}$$

QUESTION

When a clock shows 4 O'clock, how much angle do its minute and hour hand make.

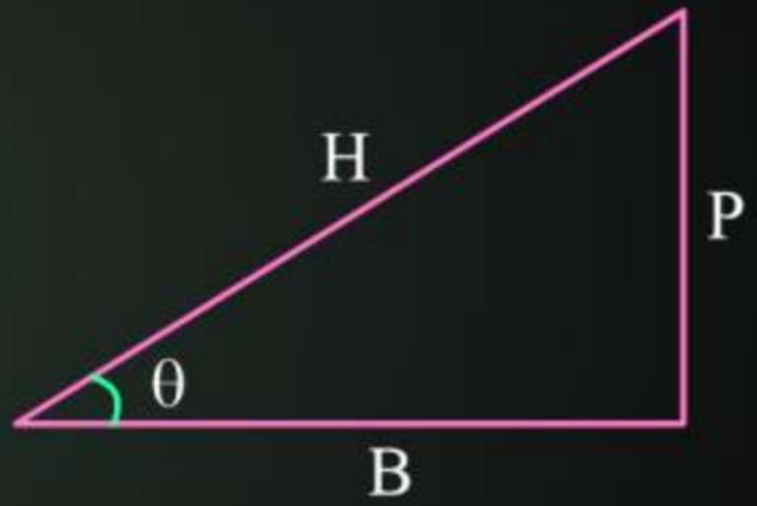


$\theta = \frac{360}{12} = 30$

$40 = \alpha$

$\frac{120}{1} = \alpha$

TRIGONOMETRIC RATIO



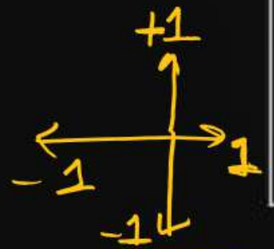
$\theta \uparrow \longrightarrow$ from $0°$ to $90°$

$\sin\theta \rightarrow$ increase, $\cos\theta \rightarrow$ decrease

$-60°$ $\quad$ $+60°$

	$0°$	$30°$	$45°$	$60°$	$90°$	$120°$	$135°$	$150°$	$180°$
$\sin\theta$	0	$\frac{1}{2}=0.5$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$+\frac{\sqrt{3}}{2}$	$+\frac{1}{\sqrt{2}}$	$+\frac{1}{2}$	0
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan\theta = \frac{\sin\theta}{\cos\theta}$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0

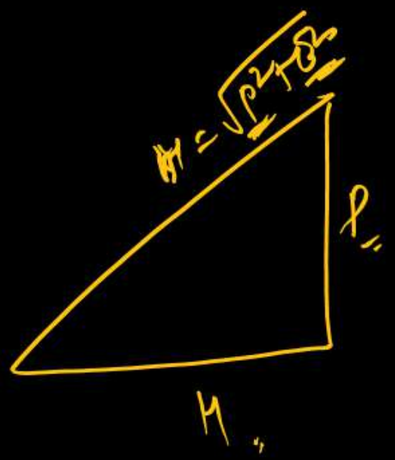
30° → 150°
45° → 135°
60° → 120°



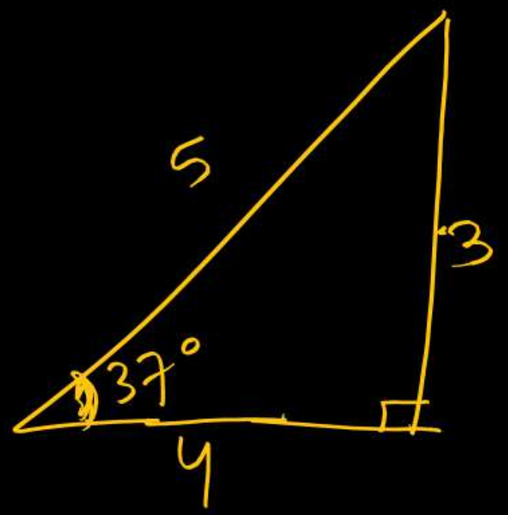
$\sin(270°) = -1$

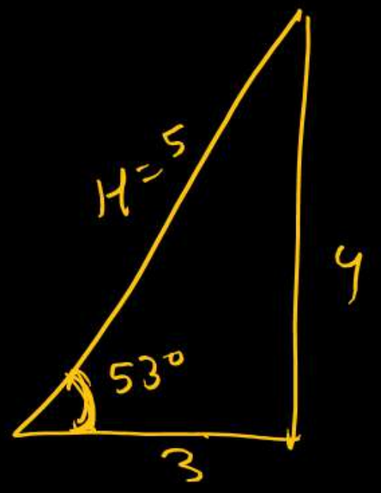
$\{-1 \le \sin\theta \le +1\}$ $\{-1 \le \cos\theta \le +1\}$ $\quad -\infty \le \tan\theta \le +\infty$

$$\sin\theta = \frac{P}{H}$$









$\sin 37° = \frac{3}{5}$

$\cos 37° = \frac{4}{5}$

$\tan 30 = \frac{3}{4}$

$\sin 53° = \frac{4}{5}$

$\cos 53° = \frac{3}{5}$

$\tan 53° = \frac{4}{3}$ ✓

QUESTION

If angle increases from $0°$ to $90°$, then value $\sin\theta$ and $\cos\theta$

A Increases, decreases

B decreases, Increases

C both decrease

D both Increase

QUESTION

If value of $\sin\theta = \frac{3}{4}$, then find value of $\cos\theta$ and $\tan\theta$.

$\sin\theta = \frac{3}{4} = \frac{P}{H}$

$P = 3$

$H = 4$

$B = \sqrt{H^2 - P^2}$

$= \sqrt{16 - 9}$

$= \sqrt{7}$

$\cos\theta = \frac{B}{H}$

$= \frac{\sqrt{7}}{4}$

$\tan\theta = \frac{P}{B} = \frac{3}{\sqrt{7}}$

QUESTION

Find value of $\sin(53°) \times \operatorname{cosec}(53°)$ = 1

QUESTION

Find value of $\sin^2(40°) + \cos^2(40°)$ = 1

$\sin^2(22°) + \cos^2(22) = 1$

$P^2 + B^2 = H^2$

divide by P^2 both side

$\frac{P^2}{P^2} + \frac{B^2}{P^2} = \frac{H^2}{P^2}$

$1 + \cot^2\theta = \operatorname{cosec}^2\theta$

divide by H^2 both side

$\frac{P^2}{H^2} + \frac{B^2}{H^2} = \frac{H^2}{H^2} 1$

* $\sin^2\theta + \cos^2\theta = 1$

$\cos\theta = \sqrt{1 - \sin^2\theta}$

$\sin\theta = \sqrt{1 - \cos^2\theta}$

divide by B^2.

$\tan^2\theta + 1 = \sec^2\theta$

QUESTION

Find value of $\sin(180°) \cdot \cos(135°) \cdot \tan(120°)$

$\sin(180^\circ) = 0$

$-\cos(45^\circ) = -\frac{1}{\sqrt{2}}$

$-\sqrt{3}$



If $\sin\theta = \frac{4}{3}$ then find $\cos\theta$, $\tan\theta$??

Ans → Not Possible

Why Why ??

Not Possible because

$\sin\theta = 3/2$ ✗

$\sin\theta \to 3$ ✗

$\sin\theta \geq 2$.

QUESTION

For given triangle, Find value

$\sin\theta = \frac{P}{H} = \frac{6}{10}$

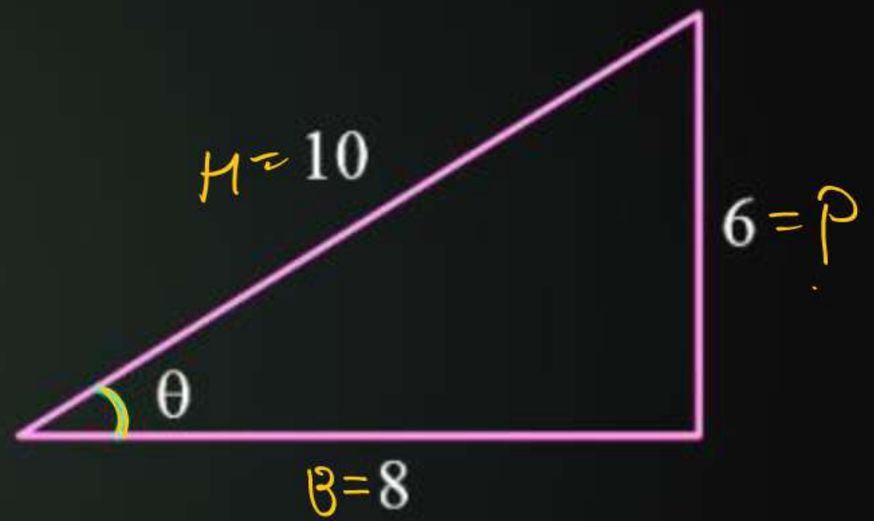
$\cos\theta = \frac{B}{H}$

$\tan\theta = \frac{P}{B}$

$\text{cosec}\,\theta = \frac{1}{\sin\theta} = \frac{10}{6}$

$\sec\theta = \frac{1}{\cos\theta}$

$\cot\theta = \frac{1}{\tan} = \frac{B}{P}$



QUESTION

If $\tan\theta = 2$ then find $\sin\theta$ and $\cos\theta$.

$\tan\theta = \frac{2}{1} = \frac{P}{B}$

$P = 2$

$B = 1$

$H = \sqrt{P^2 + B^2}$

$= \sqrt{5}$

QUESTION

If $\sin\theta = 3/5$ then find $\cos\theta$, $\tan\theta$.

$$\sin\theta = \frac{3}{5} = \frac{P}{H}$$

QUESTION

If $y = \frac{\sin\theta}{\theta}$ then find value of y if $\theta = 30°$.

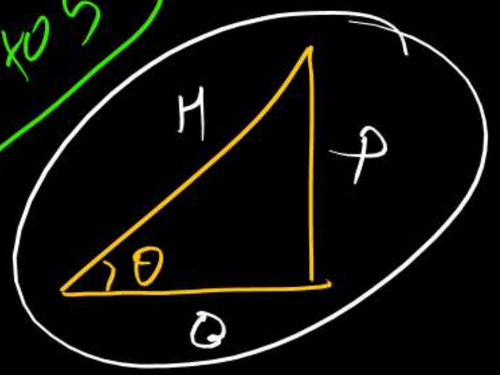
$\pi/6$ rad

$$y = \frac{\sin\theta}{\theta} = \frac{\sin 30°}{30°} \times$$

$$y = \frac{\sin 30°}{\frac{\pi}{6}\text{ rad}} = \frac{1 \times 3}{2 \times \pi} = \frac{3}{\pi}$$

Small angle approximation

up to 5°



If $\theta \to$ very small

$$\cos\theta = \frac{B}{H} = \frac{B}{B} = 1$$

$$\boxed{\cos\theta = 1}$$

$\cos 0^\circ = 1$

$\cos 1^\circ = 1$

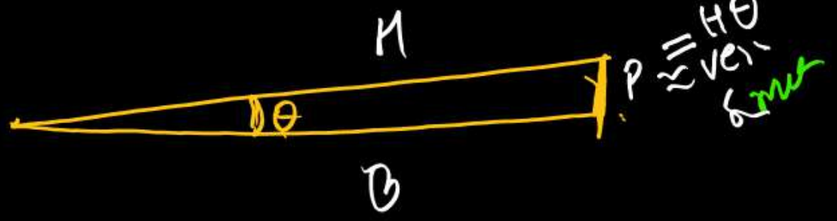
$\cos 2^\circ = 1$

$\cos 3^\circ = 1$

$\cos 5^\circ = 1$

$H = \sqrt{P^2 + B^2}$

$H = \sqrt{B^2} = B$

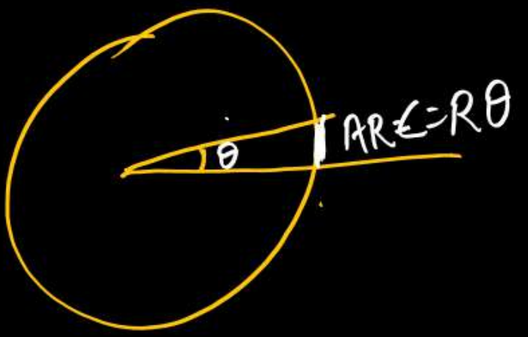


$$\# \quad \sin\theta = \frac{P}{H} = \frac{H\theta}{H} = \theta$$

θ in rad

for small angle $\tan\theta = \sin\theta = \theta$

$$\# \quad \boxed{\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\theta}{1}} \quad \theta \to \text{small}$$



QUESTION

Find value :

(i) $\sin 2^\circ = 2^\circ = 2 \times \frac{\pi \text{ rad}}{180} = \frac{\pi}{90} \text{ rad}$

(ii) $\tan 3^\circ = 3^\circ = 3\left(\frac{\pi \text{ rad}}{180}\right)$

(iii) $\cos 3^\circ = 1$.

(iv) $\sin(88.5^\circ) = \frac{P}{H} \approx 1$

(v) $\sin(89^\circ) \approx 1$

(#5) $\cos(2^\circ) = 1$

QUESTION

$(P^2 + B^2 = H^2)$

If divided by P^2 both size.

$$\theta \to \text{small} \quad \frac{\sin\theta}{\theta} = \frac{\theta}{\theta} = 1.$$

QUESTION

Find value

(i) $\sin(A+B) = \sin A \cdot \cos B + \sin B \cos A$

(ii) $\sin(A-B) = \sin A \cos B - \cos A \cdot \sin B$

(iii) $\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

(iv) $\cos(A-B) = \cos A \cdot \cos B + \sin A \cdot \sin B$

(v) $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$

(*) $\sin(105) = \sin(60+45) = \sin 60 \cdot \cos 45 + \cos 60 \cdot \sin 45$

$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}} = \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$

$\sin(75^\circ) = \sin(105^\circ)$

$\sin(75^\circ) = \sin(30^\circ + 45^\circ) = \sin 30 \cdot \cos 45^\circ + \sin 45^\circ \cos 30^\circ$

$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}$

$= \left(\frac{1+\sqrt{3}}{2\sqrt{2}}\right)$ ✓

$\cos(15^\circ) = \cos(60-45) =$

$\cos 60^\circ \cdot \cos 45^\circ + \sin 60 \sin 45$

$= \frac{1}{2}\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$

$= \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$

$\sin 60^\circ = \sqrt{3}/2$

$\cos 30^\circ = \sqrt{3}/2$

$$\sin(120^\circ) = \sin(90^\circ + 30^\circ) = \sin 90 \cos 30 + \cos 90 \sin 30^\circ$$

(A) (B)

$\cos 90 = 0$

$$= 1 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

$\sin 60^\circ = \frac{\sqrt{3}}{2}$

Ans.

$$\cos(150) = \cos 30 = \frac{\sqrt{3}}{2}$$

$$* \cos(90+60) = \cos 90^\circ \cos 60^\circ - \sin 90^\circ \sin 60^\circ$$

QUESTION

If ~~A = B~~ then $\sin(A + A) =$

If $A = B = \theta$

$$\sin(\theta+\theta) = \sin\theta\cdot\cos\theta + \sin\theta\cdot\cos\theta$$

\# $\sin(2\theta) = 2\sin\theta\cdot\cos\theta$

If $2\theta = \alpha$

$$\sin(\alpha) = 2\sin\left(\frac{\alpha}{2}\right)\cdot\cos\left(\frac{\alpha}{2}\right)$$

Half Angle formula

If $\theta = A = B$

$$\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$$

$$\cos(2\theta) = \cos\theta \cdot \cos\theta - \sin\theta \cdot \sin\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

If $2\theta = \alpha$

$$\cos(\alpha) = \cos^2\alpha/2 - \sin^2(\alpha/2)$$

$\sin^2\theta + \cos^2\theta = 1$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$\cos(2\theta) = (1 - \sin^2\theta) - \sin^2\theta$$

$$\cos(2\theta) = 1 - 2\sin^2\theta$$

$$2\sin^2\theta = 1 - \cos(2\theta)$$

$$\sin^2\theta = \frac{1 - \cos(2\theta)}{2}$$

$$\cos^2\theta = \frac{1 + \cos(2\theta)}{2}$$

QUESTION

Find value of

(i) $\sin(75°) = \sin(45° + 30°)$

(ii) $\sin(105°) =$

(iii) $\cos(120°) = \cos(90° + 30°)$

quadrant

I

II

III

IV

II

$\sin(90-\theta) = \sin 90^\circ\cos\theta - \cos 90^\circ \cdot \sin\theta = \cos\theta$

$\sin(90+\theta) = \sin 90^\circ\cos\theta + \cos 90^\circ \cdot \sin\theta = \cos\theta$

$\sin(180+\theta) = \sin 180^\circ \cdot \cos\theta + \cos 180^\circ \cdot \sin\theta = -\sin\theta$

$\sin(360-\theta) = \sin 360^\circ \cdot \cos\theta - \cos 360^\circ \cdot \sin\theta = -\sin\theta$

$\sin(-\theta) = -\sin\theta$

$\sin(180-\theta) = \sin 180^\circ \cdot \cos\theta - \cos 180^\circ \cdot \sin\theta = +\sin\theta$

$\sin(90-\theta) = +\cos\theta$

$\sin(90+\theta) = +\cos\theta$

$\sin(180+\theta) = -\sin\theta$

$\sin(360-\theta) = -\sin\theta$

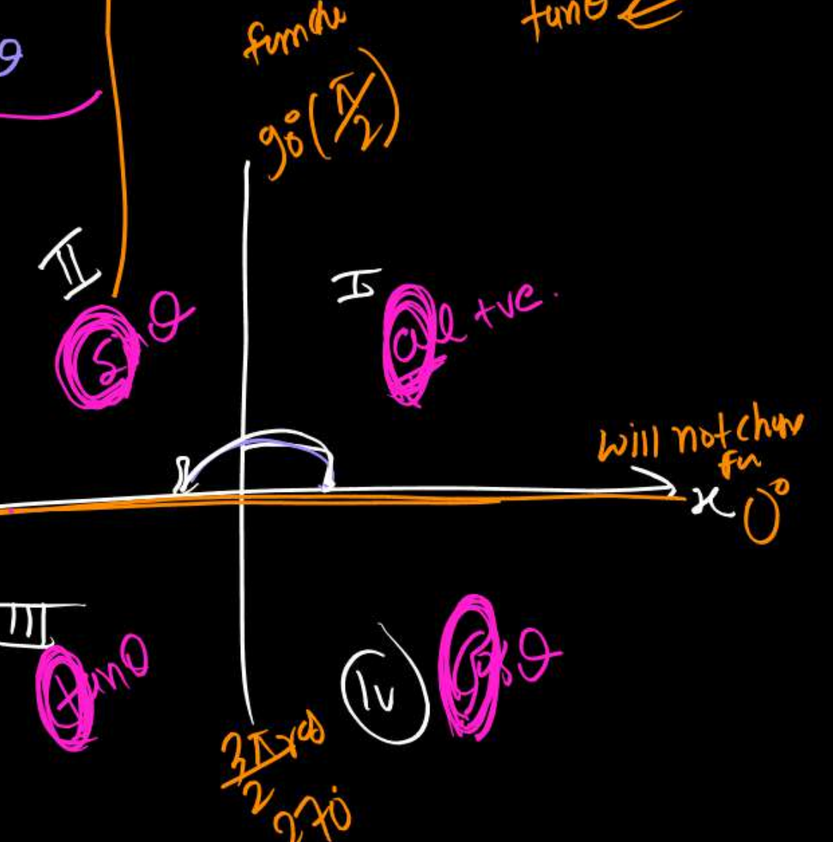
$\sin(180-\theta) = \sin\theta$

$\sin(180\pm\theta) = \mp\sin\theta$

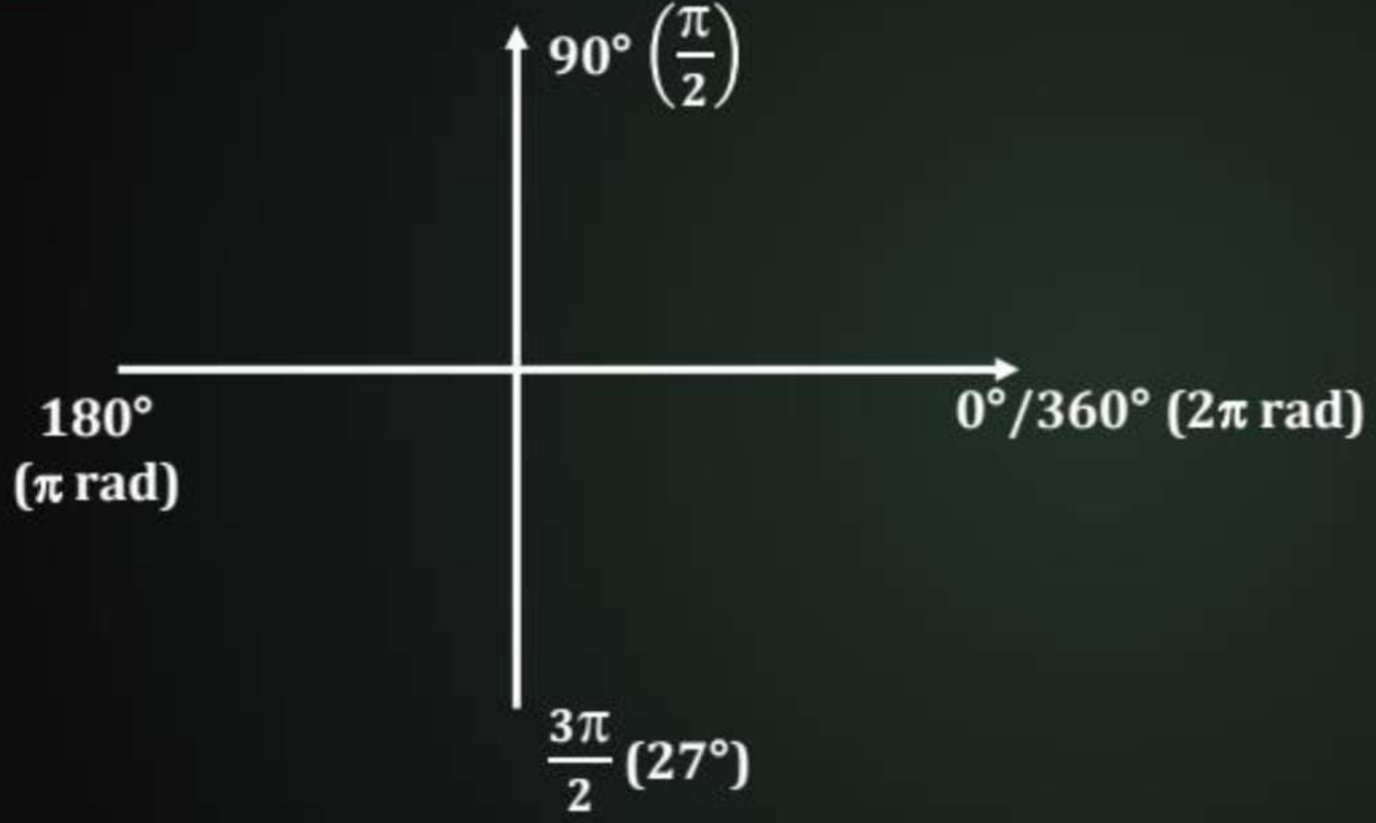
$\cos\theta(180\pm) = \cos\theta$

$\sin\theta \rightleftarrows \cos\theta$

$\tan\theta \rightleftarrows \cot\theta$



TRIGONOMETRY FUNCTION CHARGE



$\sin(90° - \theta) = +\cos\theta$

$\cos(90° - \theta) = +\sin\theta$

$\sin(90° + \theta) = +\cos\theta$

$\cos(90° + \theta) = -\sin\theta$

$\sin(180° - \theta) = +\sin\theta$

$\cos(180° - \theta) = -\cos\theta$

QUESTION

$\sin(-\theta) = \sin(0^\circ - \theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$ ✓

$\tan(-\theta) = -\tan\theta$

Find value of

(i) $\sin(-30^\circ) = -\sin 30^\circ = -\frac{1}{2}$

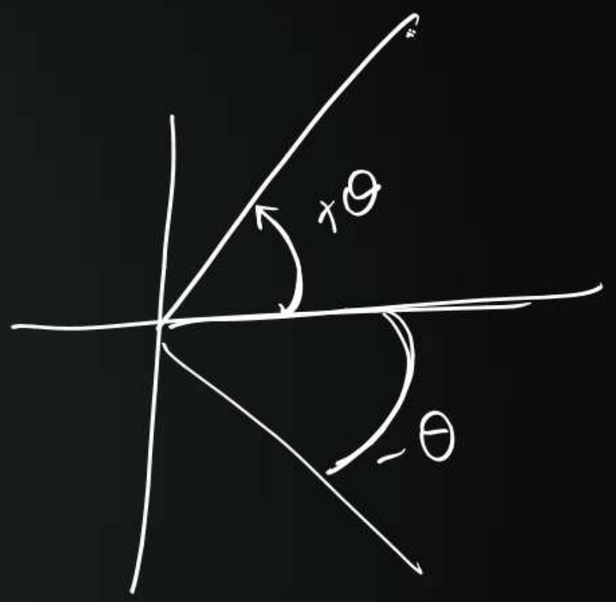
(ii) $\cos(-60^\circ) = \cos 60^\circ = \frac{1}{2}$

(iii) $\sin(120^\circ) = \frac{\sqrt{3}}{2} = \sin(90+30) = \cos 30^\circ$

(iv) $\sin(390^\circ) = \sin(360^\circ + 30^\circ) = \sin 30^\circ = \frac{1}{2}$

(v) $\sin(2\pi/3)$

✓ $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$



$\cos(-\theta) = \cos(0-\theta) = \cos 0^\circ \cdot \cos\theta + \sin 0^\circ \sin\theta$

$= 1 \cos\theta$

$\cos(-\theta) = +\cos\theta$

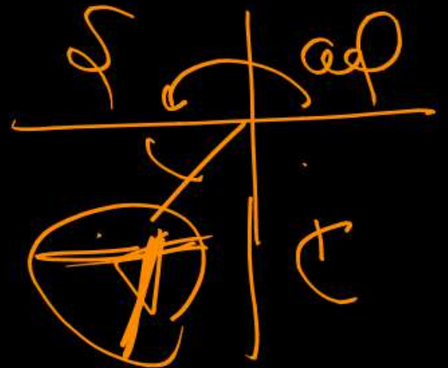
$\frac{\pi}{2}$ rad | 90°

$\sin\left(\frac{\pi}{2}-\theta\right)=\cos\theta$

$\cos\left(\frac{\pi}{2}+\theta\right)=-\sin\theta$

$\sin(\pi+\theta)=-\sin\theta$

$\tan(90-\theta)=\cot\theta$



QUESTION

If $y = A \sin\theta + B \cos\theta$ then find maximum value of y.

$y = \sin\theta$

$y_{max} = 1 \quad \theta = 90°$

$y_{min} = -1 \quad \theta = 270°$

$y = \sin(2\theta)$

$y_{max} = 1$ at $2\theta = 90°$, $\theta = 45°$

$y = \sin(3\theta)$

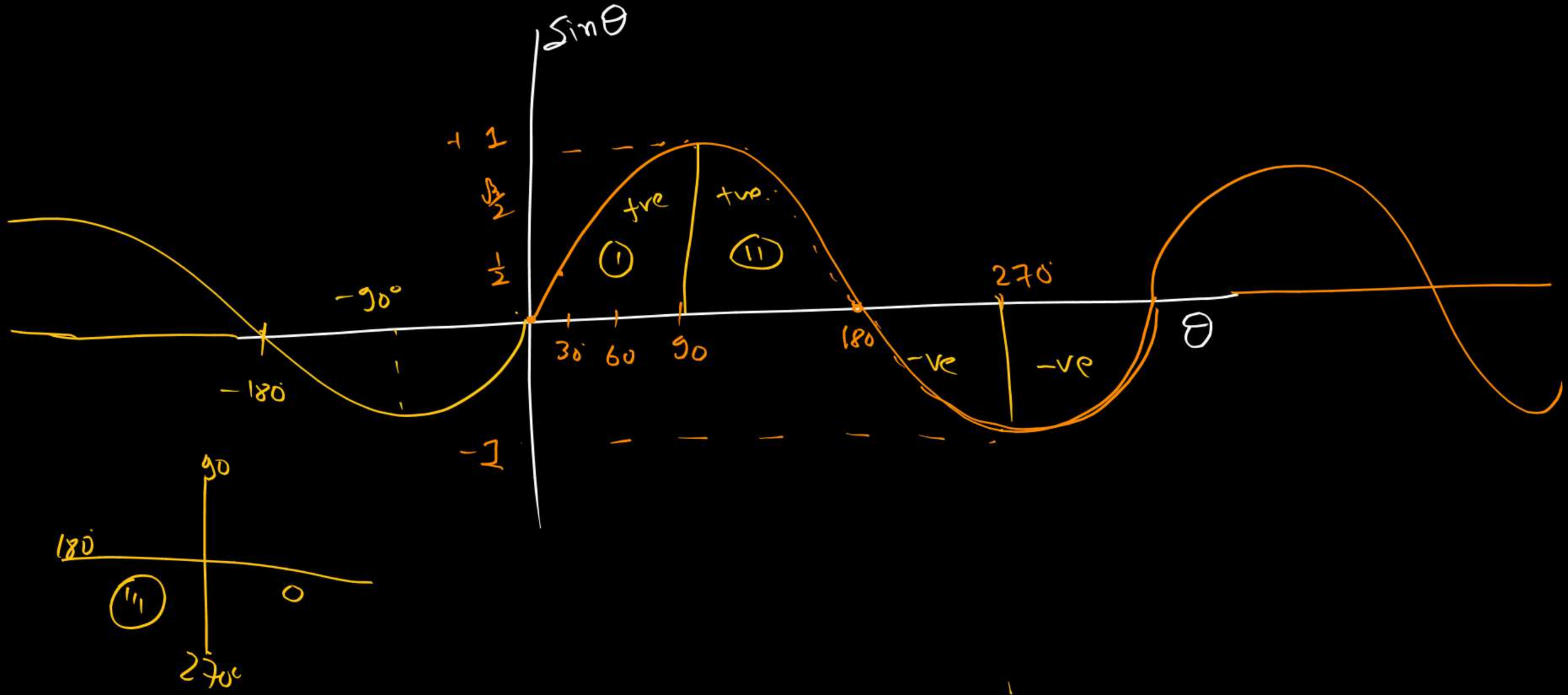
$\hookrightarrow y_{max} = 1$

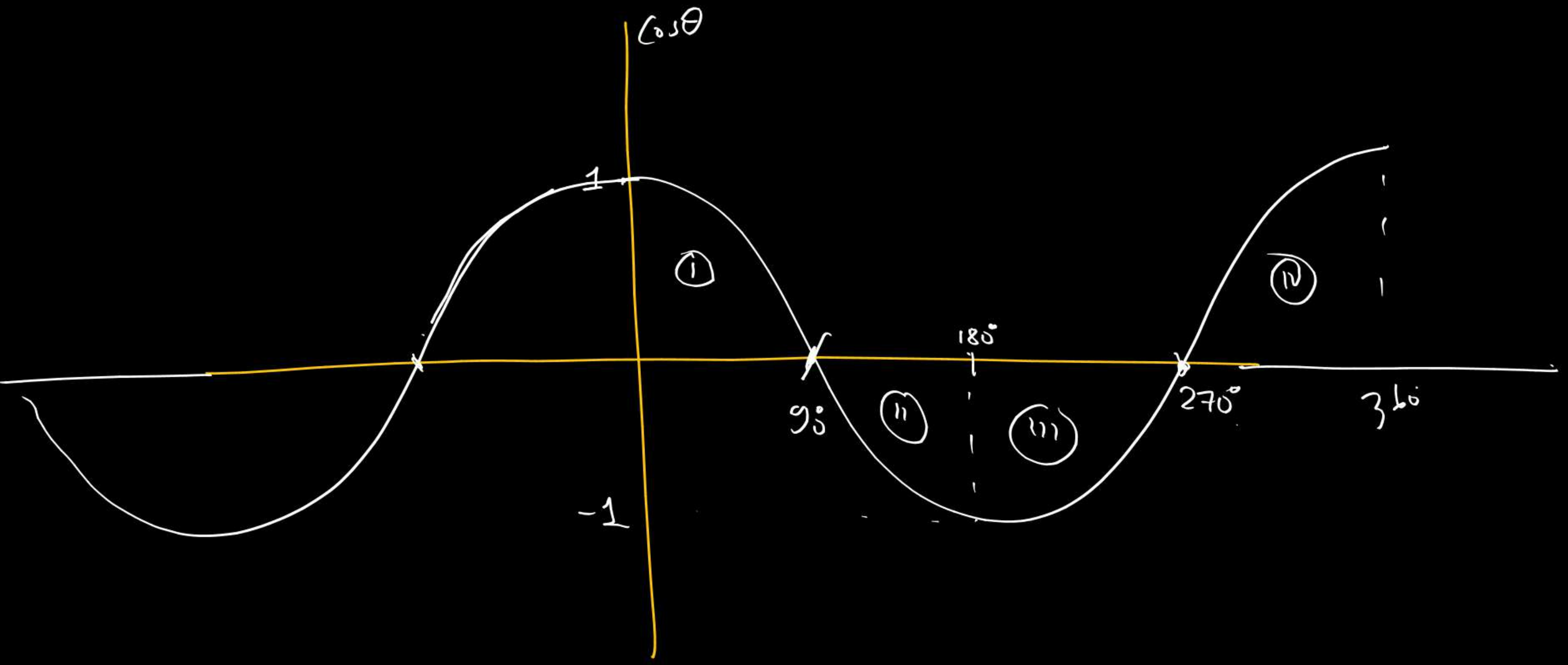
at $3\theta = 90^\circ$

$\theta = 30^\circ$

$y = 3\sin\theta$

$y_{max} = 3$


sinθ
+1
1/√2
1/2
-1
-90°
-180
30
60
90
180
270
θ
+ve
+ve
-ve
-ve
I
II
III
IV
90
180
0
270


Cosθ
1
-1
90°
180°
270°
360°
I
II
III
IV

$$Y = A\sin\theta + B\cos\theta$$

$$Y_{max} = \sqrt{A^2 + B^2}$$

(2)

$$Y = A\sin\theta + B\sin\theta$$

$$Y_{max} = \sin^{-1}(A+B)$$

$$= A+B.$$

$$y = 3\sin\theta + 4\cos\theta$$

$$y_{max} = \sqrt{3^2 + 4^2}$$

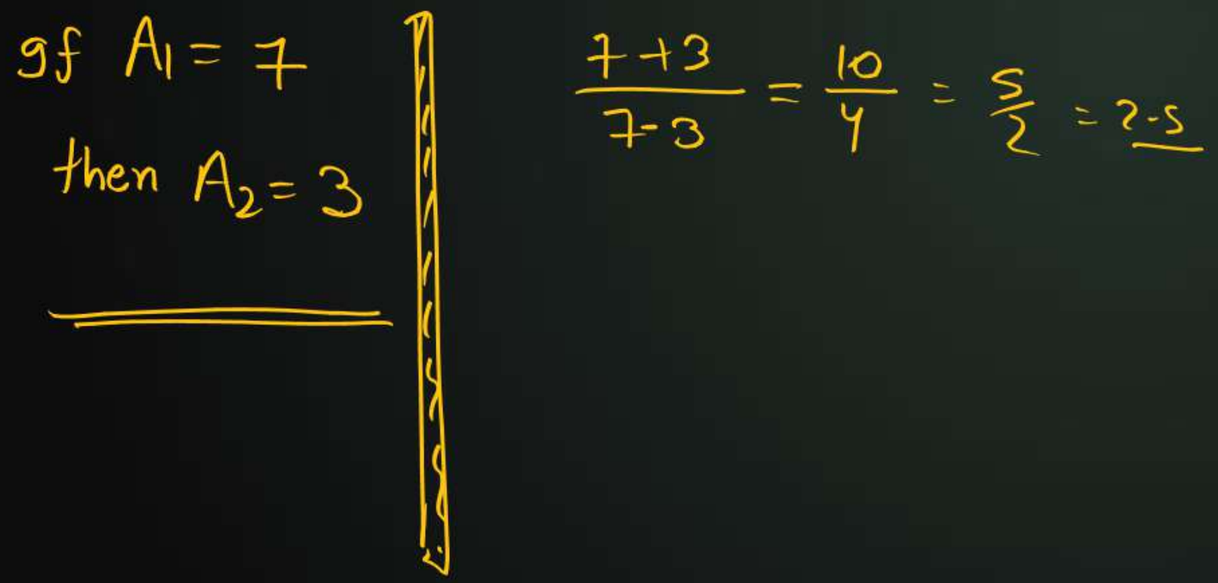
$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5$$

QUESTION

If $\frac{A_1}{A_2} = \frac{7}{3}$ then find $\frac{A_1 + A_2}{A_1 - A_2}$ = ??



$$\frac{7+3}{7-3} = \frac{10}{4} = \frac{5}{2} = 2.5$$

QUESTION

If $\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}=\frac{5}{3}$ then find $\frac{I_1}{I_2}$.

$3\sqrt{I_1}+3\sqrt{I_2}=5\sqrt{I_1}-5\sqrt{I_2}$

$8\sqrt{I_2}=5\sqrt{I_1}-3\sqrt{I_1}$

$8\sqrt{I_2}=2\sqrt{I_1}$

$\frac{\sqrt{I_1}}{\sqrt{I_2}}=\frac{8}{2}=4$

$\frac{I_1}{I_2}=16$ Ans.

QUESTION

If $\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}=\frac{5}{3}$ then find $\frac{I_1}{I_2}$.

$\sin(360°) = \sin 0°$

$\sin(450°) = \sin(360° + 90°) = \sin 90° = 1$

$\sin(-90°) = -\sin 90° = -1$

$\sin(120°) = \frac{\sqrt{3}}{2}$

$\sin(-150°) = -\sin(150) = -\frac{1}{2}$

cos (300°) =

cos (330°) = $\cos(360-30) = \cos(-30) = \cos 30 = \frac{\sqrt{3}}{2}$

tan (240°) =

cos (–30°) =

tan (–60°) =

cot (–45°) = 1

QUESTION

Suggest suitable match between function given in the first column and its description given in the second column.

Column-I	Column-II
(A) $\sin(390°) = \sin(30°) = \frac{1}{2}$	(P) Positive
(B) $\sin(-30°) = -\sin 30° = -\frac{1}{2}$	(Q) Negative
(C) $\cos 120° = -\cos 60° = -\frac{1}{2}$	(R) Zero
(D) $\tan(-120°) = -\tan(120)$ $= -(-\sqrt{3}) = +\sqrt{3}$	(S) Modulus is greater than one
	(T) Modulus is less than one

$\tan(-60) = -\tan 60°$
$= -\sqrt{3}$

$\tan(120) = -\sqrt{3}$ ✓

$\tan(-\theta) = -\tan\theta$

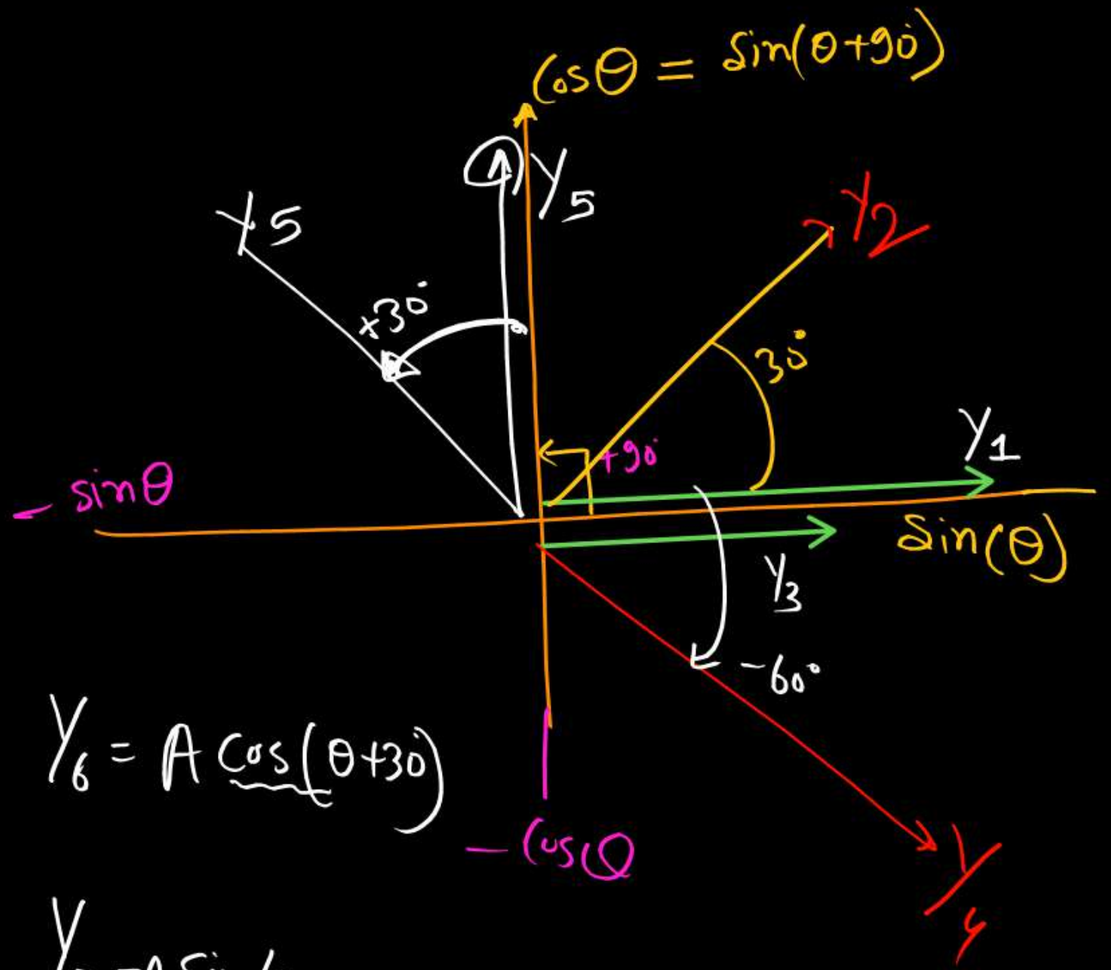
$\tan(-120) = -\tan(120)$
$= -(-\sqrt{3})$
$= +\sqrt{3}$

A. A→PT, B→QT, C→QT, D→PS

B. A→PT, B→QS, C→QT, D→PS

C. A→QT, B→QS, C→PT, D→PS

D. A→QS, B→PT, C→QT, D→PS



$Y_6 = A\cos(\theta+30)$

$Y_5 = A\sin(\theta+90) = A\cos\theta$

Phase diff^n → Angle ka diff^n

$Y_1 = A\sin(\theta)$

$Y_2 = A\sin(\theta+30)$

$Y_3 = B\sin\theta$

$Y_4 = A\sin(\theta-60)$

Phase is same for Y_1 & Y_3

QUESTION

$\theta + \frac{\pi}{4} - \theta + \frac{\pi}{4} = \frac{\pi}{2}$

Phase difference → difference in angle between two same trigonometric function. Find phase difference between A & B.

(i) $A = 4 \sin(\theta + \pi/4)$ ✓

(ii) $A = 4 \sin(\theta + \pi/3)$

(iii) $A = 3 \cos(\omega t + \pi/2)$

(iv) $A = 3 \sin(\omega t)$

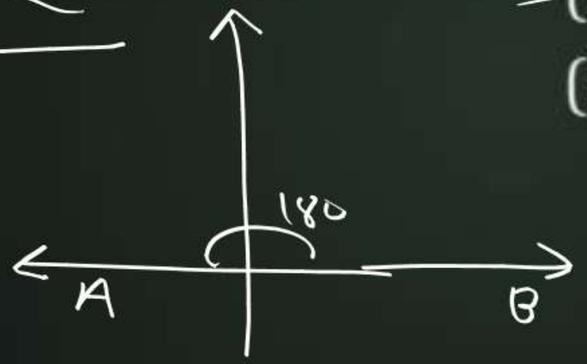
(i) $B = 3 \sin(\theta - \pi/4)$

(ii) $B = 3 \sin(\theta + \pi/6)$ $\left(\frac{\pi}{6} = \phi\right)$

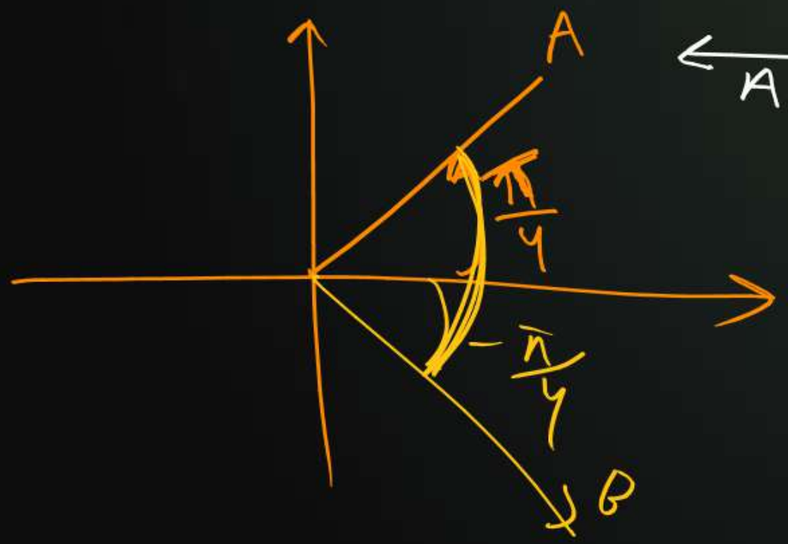
(iii) $B = 3 \sin(\omega t)$

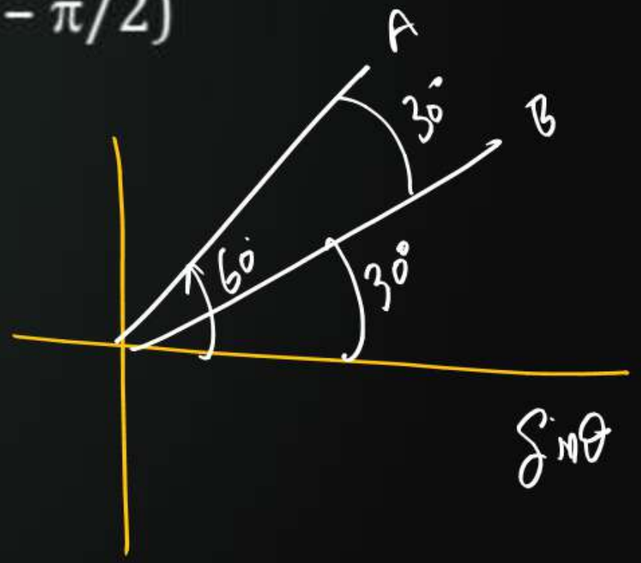
(iv) $B = 3 \cos(\omega t - \pi/2)$

$\phi = 180°$



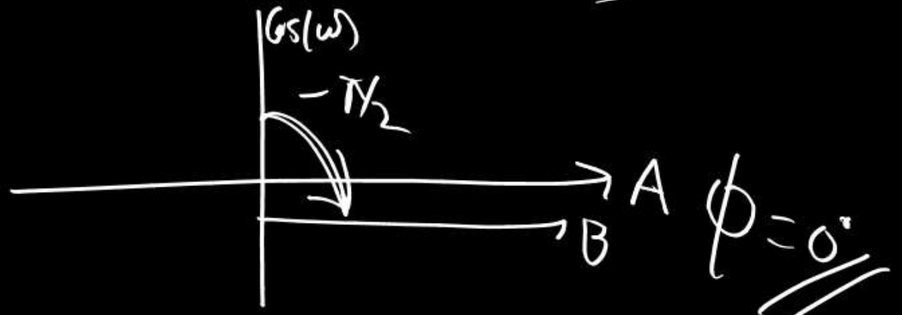
$\phi = 0°$





(iv) $A = 3\sin\omega t$

$B = 3\cos(\omega t - \pi/2)$

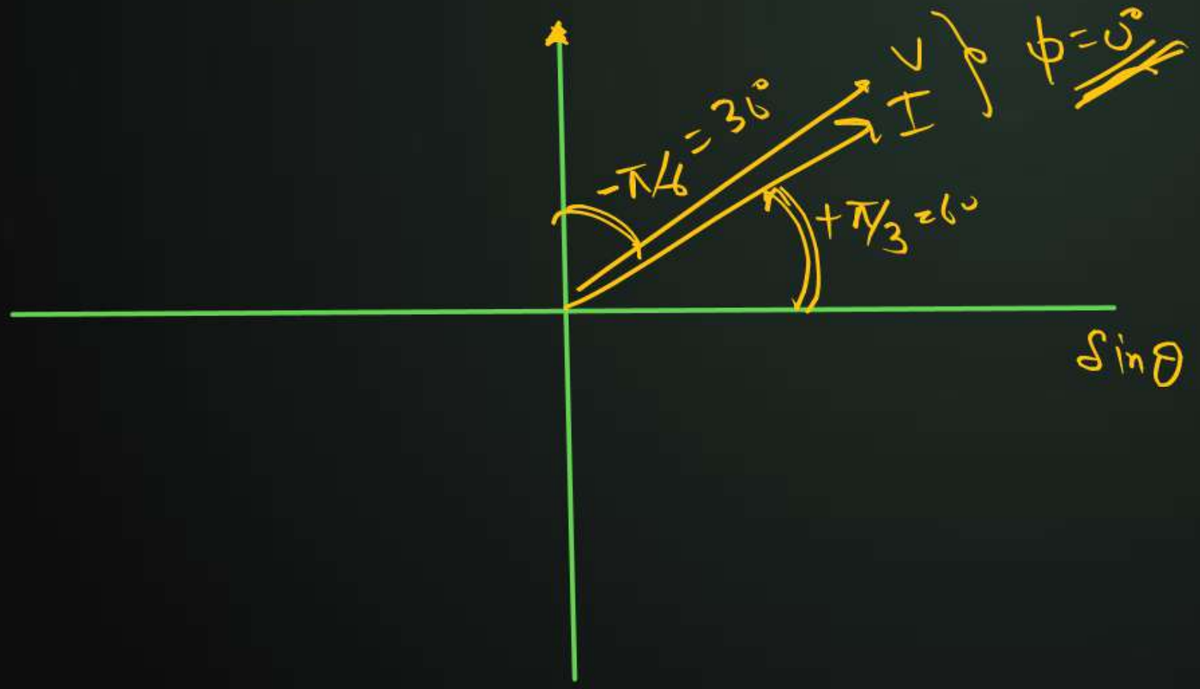


$\phi = 0°$

Equation of current and voltage

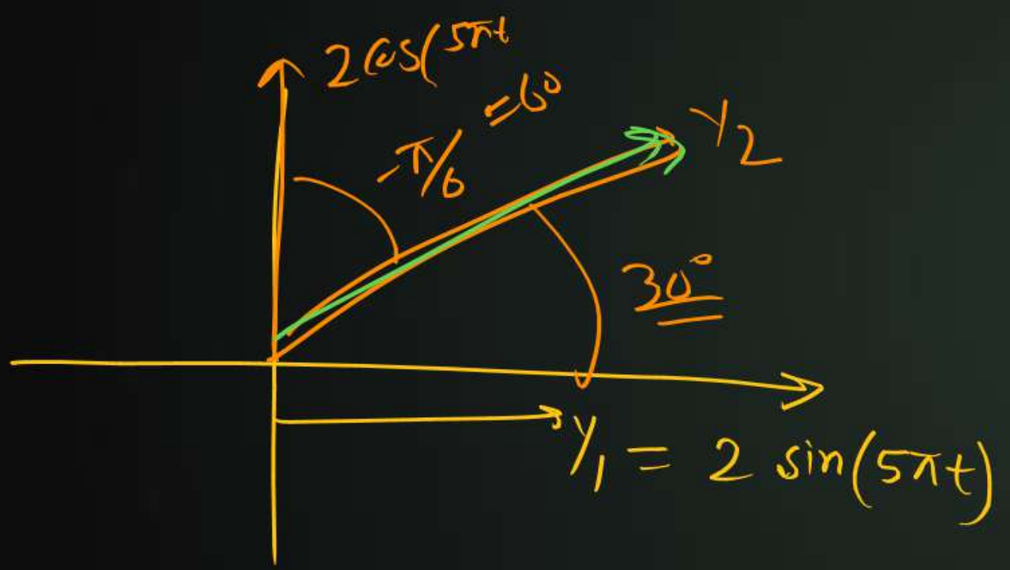
$$I = 10 \sin\left(\theta + \frac{\pi}{3}\right) \text{ and } V = 10 \cos\left(\theta - \frac{\pi}{6}\right)$$

then phase difference between current and voltage.



QUESTION

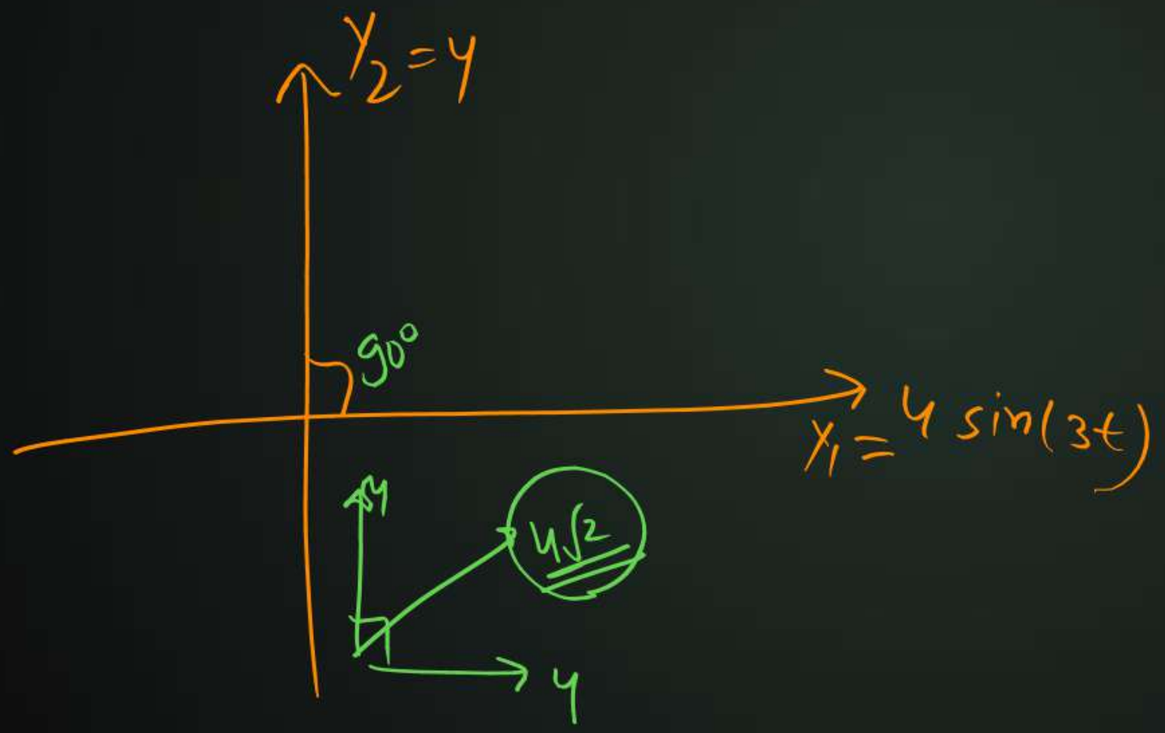
If $y_1 = 2 \sin (5\pi t)$ and $y_2 = 2 \cos (5\pi t - \pi/3)$, what is the phase difference between the two waveforms ?



$\phi = 30°$

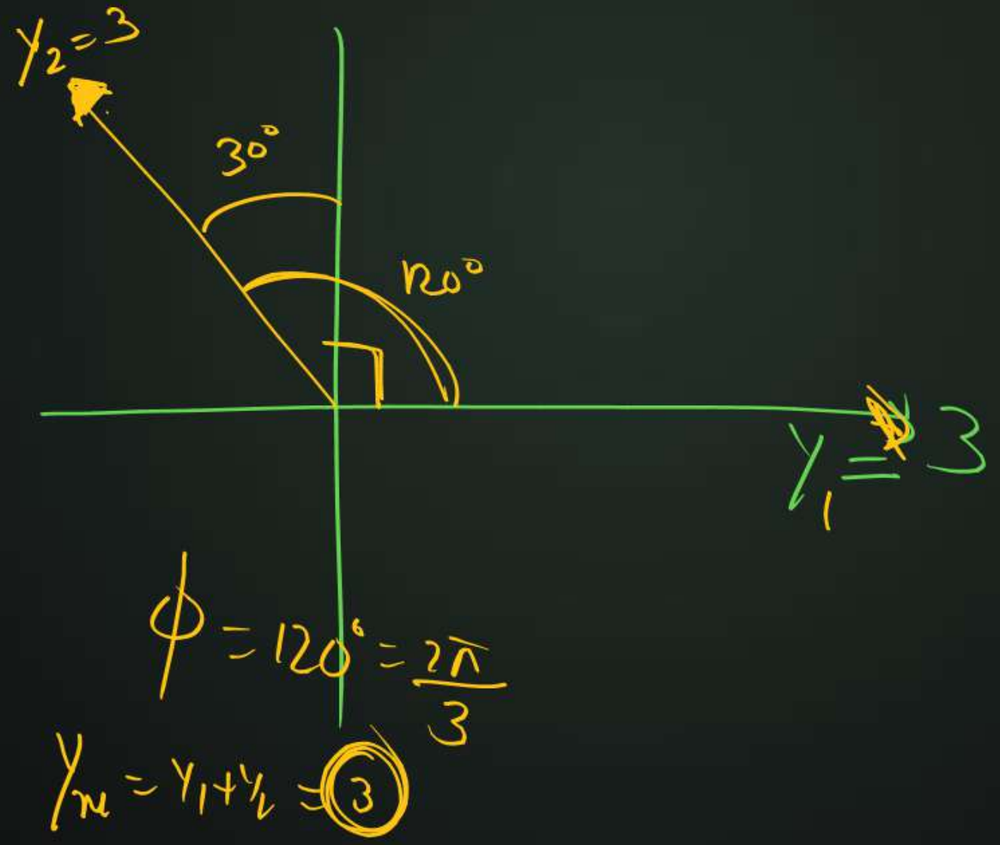
QUESTION

Two waves are represented by the equations $y_1 = 4 \sin (3t)$ and $y_2 = 4 \sin (3t + \pi/2)$. Determine the phase difference between the two waves.



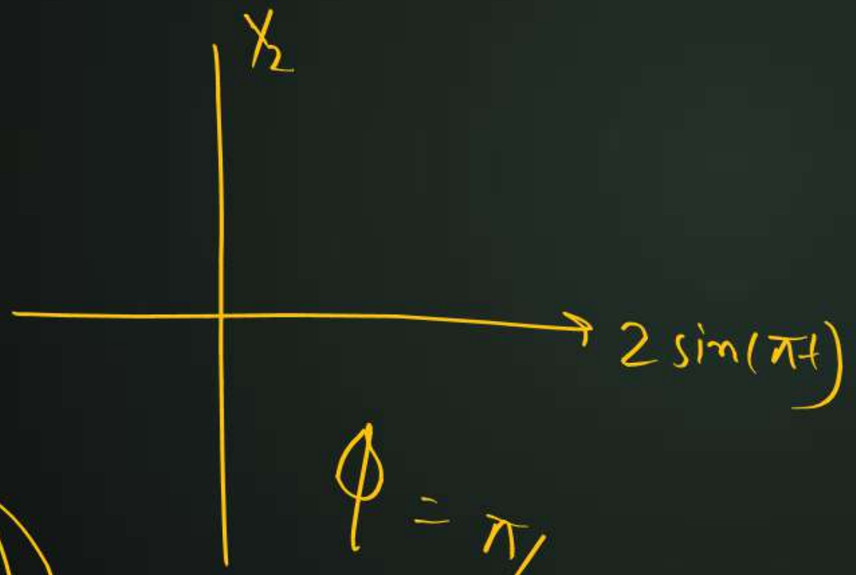
QUESTION

The equation of two waves are given as $y_1 = 3 \sin(4\pi t)$ and $y_2 = 3 \cos(4\pi t + \pi/3)$. Determine the phase difference between the two waves.



QUESTION

Two harmonic functions are described by $y_1 = 2 \sin(\pi t)$ and $y_2 = 2 \sin(\pi t + \pi/2)$. What is the phase difference between the two functions ?

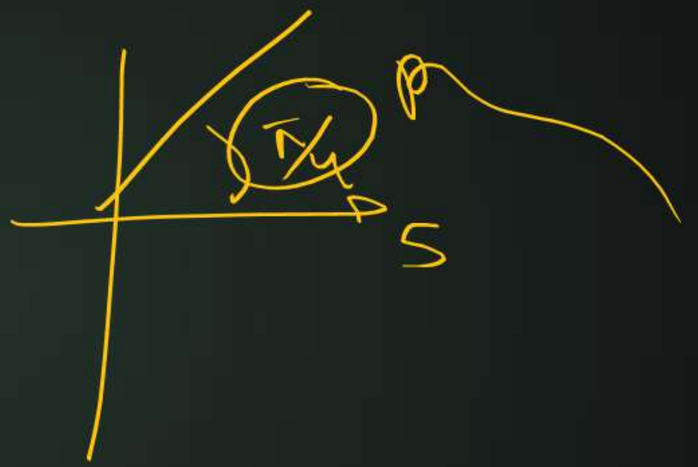


$4 - 3 = 1$

$3 - 4 = -1$

QUESTION

Two sinusoidal functions are given by $f_2(t) = 5 \sin(2\pi t)$ and $f_2(t) = 3 \sin(2\pi t + \pi/4)$. What is the phase difference between the two functions ?



QUESTION

If $y_1 = 4 \sin(\omega t)$ and $y_2 = 3 \sin(\omega t + \pi/3)$ represent two harmonic motions, what is the phase difference between them ?

$\pi/3$

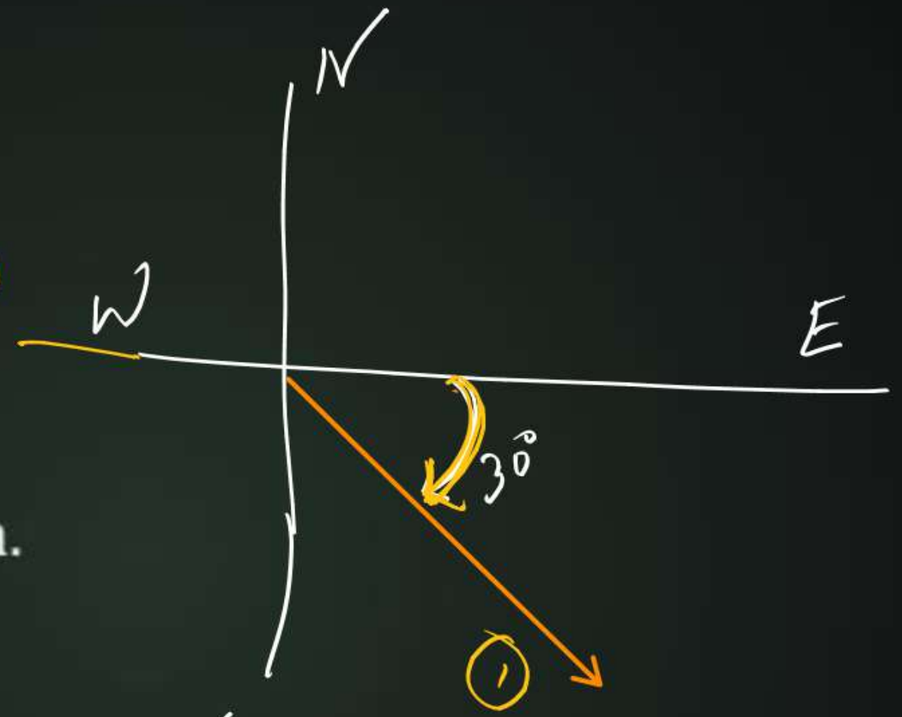
Feel of Direction

⇒ Velocity along 30° south of east

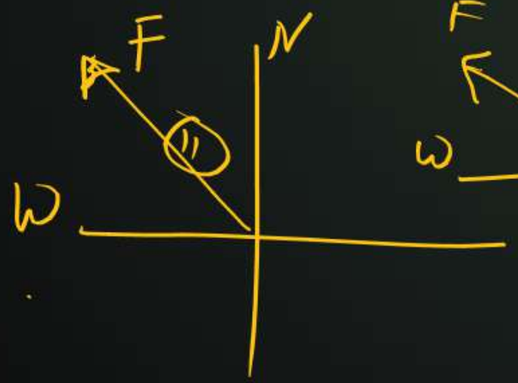
⇒ Force along North-West

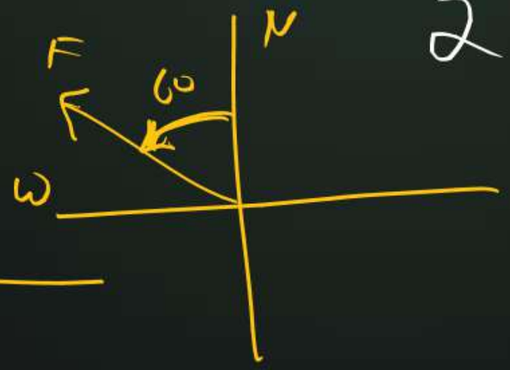
⇒ Force is acting 60° West of North.

⇒ Force along east –30° North





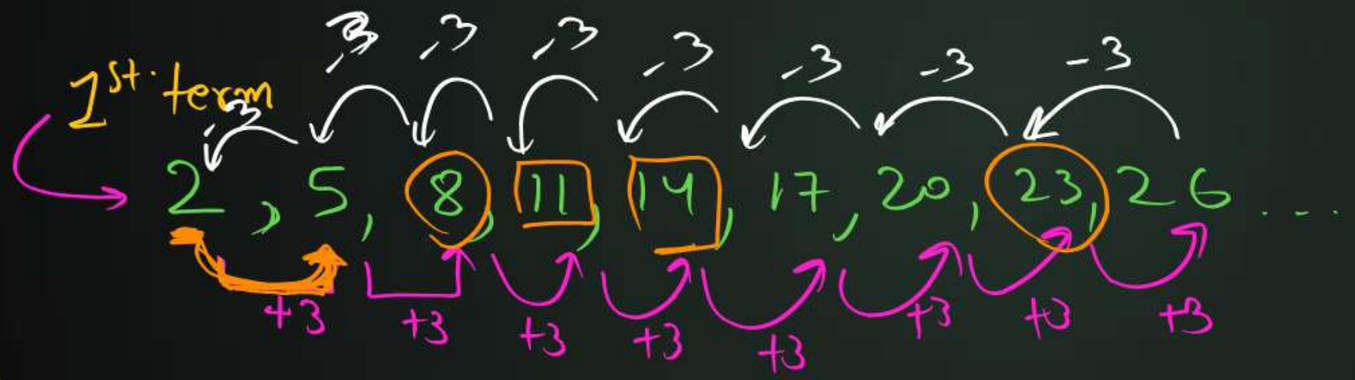




ARITHMETIC PROGRESSION

AP series → A sequence of number in which difference between any two consecutive number is a constant value.

n^{th} term $= a + (n-1)d$



$d\left(\text{common diffn}\right) = n^{th}\text{ term} - (n-1)^{th}\text{ term}$

value of nth term $= a + (n-1)d$

Sum of nth term $= \left(\frac{1^{st}\text{ term} + n^{th}\text{ term}}{2}\right)n$

$= \left(\frac{a + a + (n-1)d}{2}\right)n$

$= \frac{n}{2}\left[2a + (n-1)d\right]$

4, 7, 10, 14, 17, 20 - -

No A|p

2, 3, 5, 7, 8, 11

4, 6, 8, 10, 13, 16, 19, 22

A|P

QUESTION

Find 10th term and sum of 1st 10 term.

7, 11, 15, 19, _, _, _, _, _, _

a = 1st ten

$d = 15 - 11 = 4$

$n^{th}\text{term} = a + (n-1)d$

$= 7 + (10-1)4$

$= 7 + 9 \times 4$

$= 7 + 36 = 43$

Sum of 10 ten $= \left(\frac{7 + 93}{2}\right) \times 10$

$= 50 \times 5$

$= 250$

QUESTION

Find 51th term and sum of 51 term.

1, 3, 5, 7, 9, 11, __, __, __, __

$a = 1$

$d = 2$

5th term $= 1 + (51-1)2$

$= 1 + 50 \times 2$

$= 101$

$= \left(\frac{1st\ term + nth\ term}{2}\right) \times n$

$Sum\ of\ 51\ ter = \frac{n}{2}\left[2a + (n-1)d\right]$

$= \frac{51}{2}\left[2 \times 1 + (51-1) \times 2\right]$

$= \frac{51}{2}\left[2 + 100\right] = \frac{51 \times 102}{2}$

$= 51 \times 51$

QUESTION

Sum of 1st n-natural number $1 + 2 + 3 + \ldots\ldots\ldots n$.

A $\frac{n^2}{2}$ ✗

B $\frac{n(n+1)}{2}$ ✓

C $\frac{n(n-1)}{2}$ ✗

D $\frac{n^2(n+1)}{6}$ ✗

$a = 1$, $d = 1$

Sum of n terms $= \frac{n}{2}[2a + (n-1)d]$

$= \frac{n}{2}[2 \times 1 + (n-1) \times 1]$

$= \frac{n}{2}(2 + n - 1)$

$= \frac{n}{2}(n+1) = \frac{n(n+1)}{2}$

If $n = 1$

Sum = 1

If $n = 2$

$1 + 2 = 3$

QUESTION

28, 22, x, y, 4 is an AP then value of x and y is :

$d = n^{th} - (n-1)^{th}$

$= 22 - 28 = -6$

28, 22, 16, 10, 4

-6 -6 -6 -6

$d = -6$

QUESTION

25 tuning fork are arranged in increasing order of frequency. Frequency difference is 10 then find frequency of 1st tuning fork if frequency of 25th is 400 Hz.

$d = 10$

25th term $= 400$

n^{th} term $= a + (n-1)d$

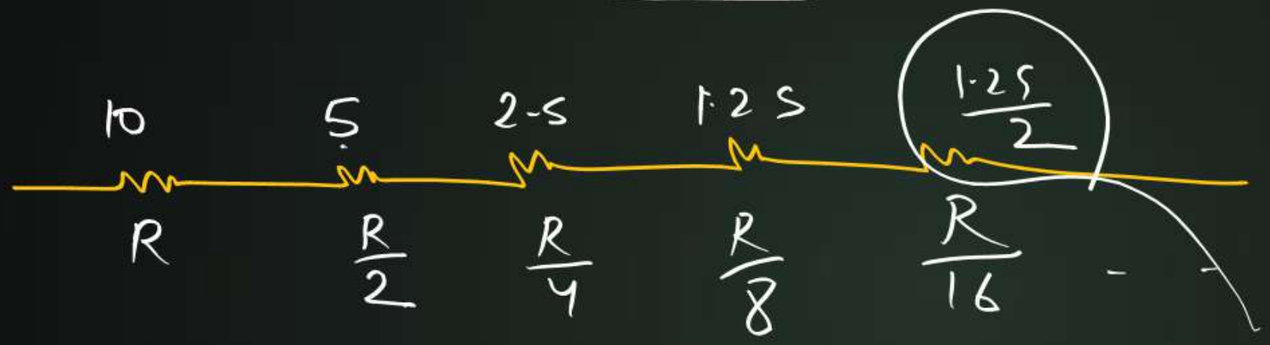
$400 = a + (24) \times 10$

$400 = a + 240$

$a = 400 - 240 = 160$

QUESTION

5-resistance connected in series, Value of resistance of each Resistor is half of previous resistance. if 1st resistance of value 10Ω then value of 5th resistance.



QUESTION

Ten tuning forks are arranged in increasing order of frequency in such a way that any two consecutive tuning forks produce 4 beats per second. The highest frequency is twice that of the lowest. Possible highest and lowest frequencies (in Hz) are

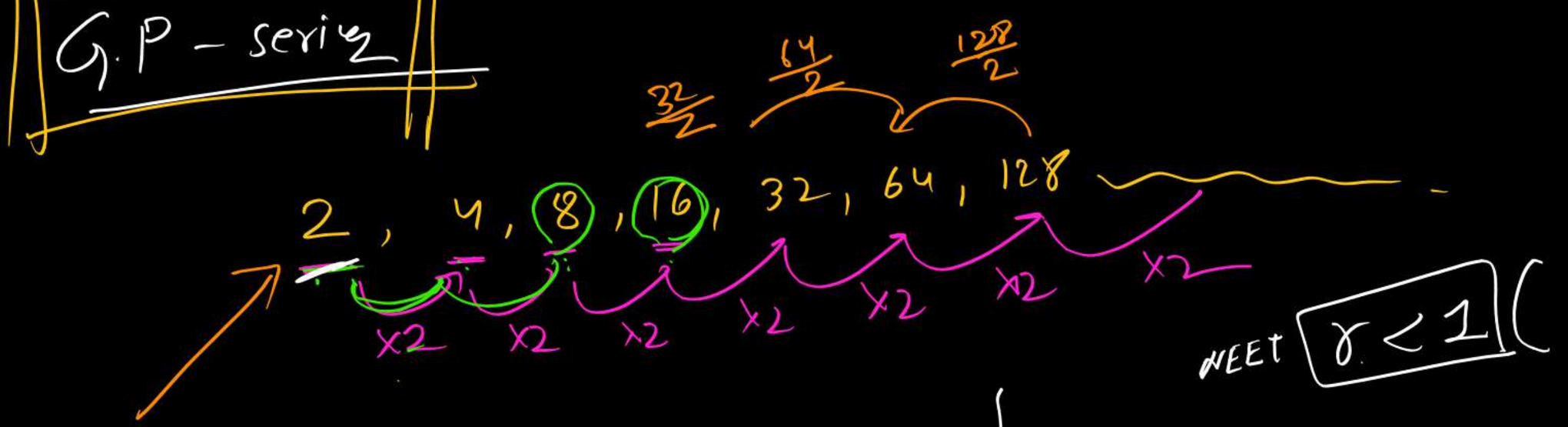
A 80 and 40

B 100 and 50

C 44 and 22

D 72 and 36

G.P - series



NEET $r < 1$ (

1st term.

r (Comm ratio) $= \frac{n^{th}\ term}{(n-1)^{th}\ term} = \frac{16}{8} = 2$

$\# \ n^{th}\ term = ar^{(n-1)}$

$Sum\ of\ all\ the\ term = \frac{a}{1-r}$

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} \cdots = 2$$

$\rightarrow$ $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16} \cdots$

1st term

$$r = \frac{\left(\frac{1}{8}\right)}{\frac{1}{4}} = \frac{1}{8} \times \frac{4}{1} = \frac{1}{2}$$

Sum of infinite term $= \frac{1}{1 - \frac{1}{2}} = \frac{1}{\left(\frac{1}{2}\right)} = 2$

Soln → $1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, -\frac{1}{32}$ ----

$a = 1$

$d = \frac{-\frac{1}{8}}{\frac{1}{4}} = -\frac{1 \times 4}{8} = -\frac{1}{2}$

Sum of ∞ $= \frac{1}{1-\left(-\frac{1}{2}\right)} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$

$1, \frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \frac{1}{256}$ ---

$a = 1 \quad r = \frac{1}{4}$

$Sum = \frac{1}{1-\frac{1}{4}} = \frac{1}{3/4}$

$= \frac{4}{3}$

QUESTION

Find potential at origin ??

0	+q	−q	+q	−q	+q	−q
(0, 0)	(1, 0)	(2, 0)	(4, 0)	(8, 0)	(16, 0)	(32, 0)

$$V = \frac{Kq}{1} - \frac{Kq}{2} + \frac{Kq}{4} - \frac{Kq}{8} + \frac{Kq}{16}$$

$$= Kq\left(1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} + \cdots\right)$$

$a = 1$, $r = -\frac{1}{2}$

$$= Kq\left(\frac{1}{1+\frac{1}{2}}\right) = \frac{2Kq}{3}$$

QUESTION

Arithmetic Progression – sequence of number in which difference between any two consecutive number is constant.

Ex. – Natural number, sequence of odd no.

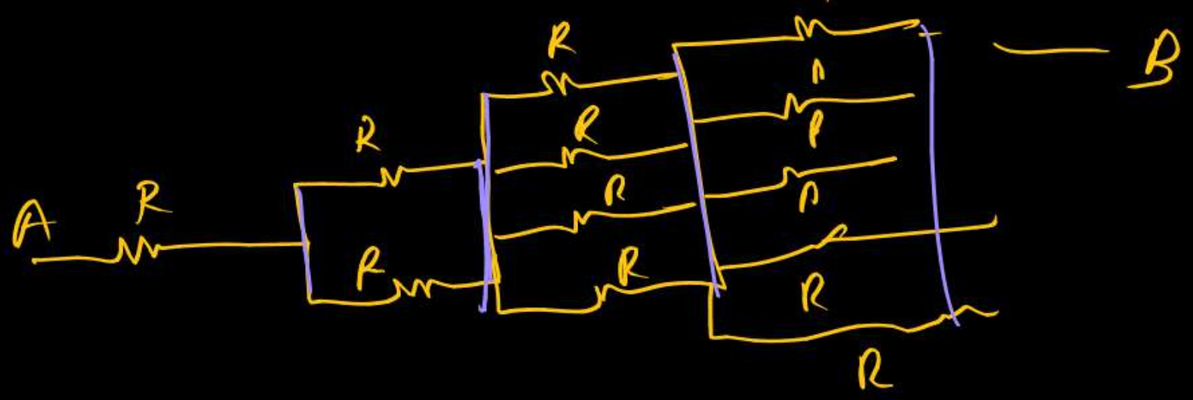
QUESTION

a = first term

d = common difference

a_n = n^{th} term

S_n = Sum of first n-term





G.P on $r > 1$

2, 8, 32, ...

Not Syllabus

$$R_{eq} = R + \frac{R}{2} + \frac{R}{4} + \frac{R}{8} + \frac{R}{16}$$

$$= R\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}\right)$$

$$= R(2) = 2R$$



$F_{net\ gravitational\ force\ on\ m_0} = \frac{4Gmm_0}{3}$??

(#) Graph → 1h

Quadrat → 20
diff^n (max u → 1h
Integ → 1h
Binom → 20

THANK YOU

